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When connectivity fails: climate exposure, delivered prices and poverty risk in island regions

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ABSTRACT

Archipelagic economies face a distinct interaction between climate risk and spatial inequality, as disruptions to sea connectivity can simultaneously reduce earnings and increase the cost-of-living in isolated islands. This article establishes a climate-enhanced New Economic Geography framework in which climate exposure affects welfare through two interrelated mechanisms: a connectivity channel that raises effective trade and logistics costs, consequently increasing the local price index, and an agglomeration channel that reduces market access and local income potential in peripheral regions. The model highlights a fiscal policy dilemma within a limited budgetary context: subnational governments must allocate scarce resources among connectivity investments, resilience expenditures and social protection initiatives. A stylized quantitative calibration, implemented in a two-region core–periphery environment, shows that heightened exposure can increase poverty risk even when nominal output changes only modestly, because price differentials raise the cost of basic consumption in remote areas. The numerical results are model-implied predictions rather than estimates for a specific country, and they illustrate how combinations of policy instruments can outperform single-instrument strategies: connectivity investment reduces delivered-cost wedges, resilience lessens the severity and duration of disruptions and adaptive social protection stabilizes welfare during shocks. These predictions are intended to inform measurement and policy development in archipelagic settings, such as Indonesia.

IMPACT STATEMENT

Climate change is progressively acknowledged as a catalyst for economic disparity; nevertheless, current spatial-economic models hardly account for the unique vulnerabilities of archipelagic and small-island economies, where maritime connectedness, rather than mere distance, influences market accessibility and living expenses. This study proposes a Climate-Enhanced New Economic Geography model illustrating that climate vulnerability may contribute to poverty in remote regions via two separate mechanisms: by disrupting trade and increasing consumer prices, and by diminishing the agglomeration dynamics that support local income growth. Employing a simplified core-periphery calibration inspired by Indonesia's archipelagic landscape, the study reveals that the likelihood of poverty may increase despite relatively slight alterations in nominal income, as the living expenses in isolated islands are acutely affected by disruptions in connectivity. The importance of the work resides in redefining climate adaptation as a public finance issue instead of merely an infrastructural challenge: it demonstrates that investments in connectivity, expenditures on resilience, and social protection function through various welfare margins and yield optimal results when combined rather than separately. This is specifically related to financially limited subnational governmental entities in archipelagic countries, like the outer islands of Indonesia, where budgetary decisions within intergovernmental transfer frameworks must reconcile long-term productivity improvements with immediate consumption stabilization. By making the cost-of-living channel explicit and testable, the framework also offers a concrete empirical agenda for future work using subnational panel data on connectivity, prices, and poverty in climate-exposed island economies.

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1. Introduction

Climate change is progressively reshaping both public finance and environmental policy. This article focuses primarily on physical climate risks, particularly the disruption of coastal infrastructure, port operations and maritime connectivity, while acknowledging that transition risks are correlated and may reinforce these physical channels over time. The regional distribution of climatic vulnerability is uneven, as are the fiscal capabilities and service delivery obligations of subnational governments. When physical hazards impair transport infrastructure and raise the cost-of-living in remote areas, poverty may increase even without significant changes in nominal income. For archipelagic nations, this poses a critical policy dilemma: how should public resources be distributed among connectivity investment, resilience and social protection to prevent climate risk from entrenching long-run poverty?

Indonesia is an appropriate illustrative setting for the model for two related reasons. First, it is the world's largest archipelagic state, with more than 17,000 islands, so remoteness from the national economic core is a structural feature of many outer-island regions. This makes inter-island connectivity central to market access, delivered prices and welfare, matching the mechanisms developed in this article. Second, although Indonesia has undergone extensive fiscal decentralization, the central government continues to play a major role in financing subnational governments through intergovernmental transfers. This institutional structure makes Indonesia especially useful for illustrating how connectivity investment, resilience spending and social protection interact within a national fiscal framework, whereas Pacific Small Island Developing States (SIDS) are useful examples of remote-island vulnerability but are less suitable for examining large intergovernmental transfer systems of this kind. This emphasis does not imply that climate damages are negligible in Indonesia's major metropolitan areas. The concentration of population and economic activity in places like Jakarta means that even localized damage can generate large aggregate losses. The focus here, however, is distributional. The question is why climate risk falls disproportionately on peripheral islands relative to less exposed areas, and how public policy can reduce that disparity.

The budgetary rationale is well established within fiscal federalism: when risks and needs differ geographically, intergovernmental systems face an equalization challenge, and public spending priorities must balance efficiency with equity (Musgrave, 1959; Oates, 1972, 1999). This article applies New Economic Geography (NEG) to examine the spatial mechanisms underlying these distributional concerns. NEG provides an explanation for the disparities in economic activity between core and peripheral regions, emphasizing market access, transportation costs and agglomeration economies (Fujita et al., 1999; Fujita & Thisse, 2013; Krugman, 1991).

Recent developments in quantitative spatial economics have incorporated climate change into migration, trade, sectoral specialization and infrastructure vulnerability models, demonstrating how climate shocks can affect spatial equilibria and welfare across regions (Balboni, 2025; Conte et al., 2021; Cruz & Rossi-Hansberg, 2024; Desmet & Rossi-Hansberg, 2024; Hsiao, 2025). This article situates itself at the intersection of this spatial climate literature and the public finance problem of protecting welfare in fiscally constrained, climate-exposed peripheries. It does so through a stylized but internally consistent framework that extends a standard NEG core–periphery structure by allowing physical climate exposure to shift effective trade costs and agglomeration conditions in a transparent, reduced-form way.

The framework links climate exposure to poverty through two spatial channels. First, a connectivity channel: physical climate risk raises effective trade and logistics costs by damaging infrastructure and increasing the probability of transport disruptions, thereby reducing market access for peripheral islands. This channel increases local prices and the cost-of-living, which is central to poverty in isolated island settings where imported consumption baskets are large and substitution options are limited. Second, an agglomeration channel: heightened exposure and transport uncertainty reduce the expected profitability of locating firms and capital in vulnerable areas, weakening agglomeration economies and lowering local income potential. Together, these channels imply that poverty is affected by both income changes and cost-of-living shifts, so that climate risk can intensify poverty through price effects even when nominal output changes only modestly.

A clarification on the model implications is warranted here. The three theoretical propositions developed in Section 3 are model implications derived from sign restrictions on parameters, not empirically

tested claims derived from estimation. Specifically, they follow from the assumption that the sensitivity of trade costs and agglomeration to climate exposure is positive. These propositions frame the simulation and point toward testable empirical mappings; they are not claims about estimated coefficients.

This framework carries distinct implications for public finance. If climate exposure simultaneously reduces local income potential and raises local living costs, a single policy instrument is unlikely to be sufficient. A combination is required. Connectivity investment mitigates baseline isolation and dampens price pass-through from disruptions. Resilience expenditure protects essential coastal assets and slows the accumulation of exposure over time. Adaptive social protection stabilizes welfare when disruptions temporarily raise prices and reduce earnings. The model does not assert that any one policy universally dominates; it clarifies the conditions under which each instrument is most effective and why the optimal combination varies across island locations depending on exposure intensity, distance and baseline connectivity.

The article proceeds in two stages. First, it develops a spatial core–periphery model for an archipelagic economy in which climate exposure affects welfare through two linked channels: a production channel that lowers effective productivity and income, and a market-access channel that raises delivered prices through disrupted connectivity. The model builds on established NEG and quantitative spatial equilibrium structures, particularly Krugman (1991) and Fujita et al. (1999), and introduces climate exposure as a state variable that modifies trade costs and agglomeration in a reduced form but analytically tractable way. Second, it uses simulation to evaluate the model's implications under alternative exposure and policy scenarios. The simulation varies baseline connectivity frictions, the intensity of climate disruption and the allocation of a fixed fiscal envelope across connectivity investment, resilience spending and social protection. Outcomes are reported in terms of real income, the local price index and poverty risk derived from a consumption-based welfare mapping. Sensitivity checks assess whether results hold across plausible parameter ranges and alternative assumptions on disruption persistence.

The core questions motivating this article are: how does climate exposure change poverty risk in archipelagic economies when connectivity is both costly and unreliable? Through what mechanisms can the same climate shock raise poverty by lowering nominal incomes and by raising the local cost-of-living? And what does this imply for the composition of public spending across connectivity investment, resilience and adaptive transfers under a binding fiscal envelope?

The remainder of the article is organized as follows. Section 2 reviews related literature. Section 3 develops the conceptual framework and derives model implications. Section 4 presents the formal model and comparative statics. Section 5 discusses policy implications and testable empirical mappings. Section 6 provides calibration and simulation. Section 7 concludes.

2. Literature review

Understanding why economic activity concentrates in 'core' places while peripheral regions lag has long been a central question in regional economics and development. In archipelagic nations, such as Indonesia, the issue of connectivity is intrinsically linked to inter-island transportation costs, route reliability and access to major consumer markets, which often influence corporate location decisions and household income levels. This article resides at the convergence of (i) NEG, which elucidates core–periphery divergence via market access, increasing returns and trade costs, and (ii) public finance, where intergovernmental transfers, social protection and infrastructure expenditure serve as traditional instruments for equalization and place-based development policy.

A primary motivation for expanding NEG in small-island contexts is that 'trade costs' are not only increased but also volatile. Climate dangers disrupt port operations, coastal highways and shipping reliability, while market access become more uncertain. Recent quantitative spatial economics has started modeling climate change as a trigger for spatial reallocation via migration, trade, specialization and infrastructure vulnerability (e.g. Balboni, 2025; Conte et al., 2021; Cruz & Rossi Hansberg, 2024; Desmet & Rossi Hansberg, 2024; Hsiao, 2025). The objective of this work is to apply this logic to a distinctly archipelagic core–periphery context, whereby climate risk manifests through transportation disruptions between islands and central markets, and poverty is influenced by both income and cost-of-living factors.

2.1. Economic geography and small island development

Classical location theories (von Thünen, 1826; Weber, 1929) focus on cost-minimizing factors like transportation and input expenses, although they inadequately account for the self-reinforcing clustering that defines contemporary economies. NEG addresses this issue by integrating growing returns to scale, monopolistic competition and iceberg trade costs, resulting in 'circular causation' where market size and company entry mutually enhance one another (Fujita et al., 1999; Fujita & Thisse, 2013; Krugman, 1991). In these models, higher market access raises expected profits, attracting firms and workers, expanding local variety and further increasing real wages and demand. Contemporary research in 'modern spatial economics' reinterprets these concepts using quantitative spatial equilibrium terminology, highlighting how the convergence of spatial labor demand and supply establishes the equilibrium distribution of population and production, with market access serving as the crucial sufficient statistic linking geography to welfare and regional disparity (Allen & Arkolakis, 2023).

Such spatial dynamics are especially important in small-island and archipelagic contexts, where market access is determined not only by distance but also by the reliability and cost of marine connections. In island settings, basic trade and logistics costs may be higher because of limited scale, few backhaul opportunities and dependence on a small number of ports and routes. This can weaken agglomeration forces in peripheral regions, limit firm entry and specialization, and generate persistent real-income disparities even within a single country. Spatial economics research has shown repeatedly that temporary shocks, such as disruptions to transportation networks, can have lasting spatial effects by altering migration incentives, investment patterns or the relative attractiveness of locations (Kleinman et al., 2023; S. J. Redding, 2024).

A further aspect worthy of consideration is the role of resources and capacity constraints particular to small-island settings. Islands have structural constraints in land availability, population size and migration options that affect both the strength of agglomeration forces and the capacity to adapt to climate risk. Limited land constrains physical clustering; small population increases the per-unit cost of public infrastructure; and out-migration of skilled workers diminishes the human capital concentration that underpins knowledge spillovers and productivity growth (Duranton & Puga, 2004). Higher wages in regions attract labor inflows that stimulate further capital accumulation (Fujita et al., 1999), where population and land constraints are binding, the circular-causation mechanism is weakened, and peripheral islands are structurally less able to sustain agglomeration. In Pacific and Southeast Asian SIDS, this is further complicated by remittances and external capital flows, which can partially substitute for local income losses during climate disruptions but may also dampen incentives for local adaptation investment (Chowdhury et al., 2025). The present model does not fully address these limitations, but they are treated as implicit in the low baseline agglomeration potential ($\overline{\phi}_i$) and restricted productivity scale (A_i) of peripheral regions. Extending the framework to incorporate endogenous migration, land constraints and remittance flows is an identified priority for future structural work.

A key result of this work is that island inequality encompasses not only nominal income disparities but also spatial variations in prices and the local cost-of-living. When consumption is significantly reliant on shipped items, increased delivered-cost differentials can diminish real earnings, even in the absence of substantial alterations in nominal output. Recent research pertaining to Indonesia corroborates this mechanism: policies supporting maritime logistics are demonstrated to lower food prices, suggesting that connectivity and shipping frictions significantly influence local consumer pricing (Sofiyandi et al., 2023).

In conclusion, small islands may persist in a low-market-access equilibrium despite the presence of a 'core' in other regions of the country. This finding corresponds with empirical and policy-related research that links remoteness to decreased trade flows and lower income levels (S. Redding & Venables, 2004).

2.2. Climate change and regional economic disparities

A large climate economics literature shows that environmental conditions can affect economic outcomes not only through short-run losses, but also through persistent changes in growth paths and vulnerability. Using within-country variation over time, Dell, Jones and Olken document that higher temperatures reduce economic growth in poorer countries and can affect growth rates rather than only output levels

(Dell et al., 2012). Complementary evidence from cyclone shocks indicates that large disasters can have long-run impacts on economic growth, consistent with the idea that repeated climate events can shift development trajectories (Hsiang & Jina, 2014). These findings are directly relevant to small islands because exposure is often concentrated in coastal infrastructure and climate-sensitive livelihoods, while adaptation capacity is uneven across regions.

Recent quantitative spatial economics pushes this logic further by modeling climate change as a spatial reallocation problem. In these frameworks, warming changes the relative attractiveness of locations, and the aggregate cost of climate change depends on spatial frictions, such as mobility, trade costs and the ability to reoptimize where people live and produce. Cruz and Rossi-Hansberg (2024) show how global warming can reshape the spatial distribution of economic activity and welfare in a spatial equilibrium setting. Related syntheses highlight that climate damage is not regionally uniform and that policy-relevant implications rely on the interplay between local exposure, adaptation capacities and spatial connections. In archipelagic economies, connectivity is the essential spatial linkage: when climate stress impacts ports, maritime routes and logistical reliability, the welfare consequences can be exacerbated in peripheral islands due to disruptions in both production capabilities and access to diverse resources from the core.

A secondary and frequently undervalued channel operates through pricing and the cost-of-living. The danger of poverty depends on purchasing power rather than solely on nominal income; thus, variations in geographic pricing can alter welfare assessments and poverty metrics, even when nominal wages appear comparable. This is particularly crucial for remote areas that rely on imports and experience greater delivered-cost disparities during outages. Recent World Bank research suggests enhanced techniques for calculating spatial price indices and emphasizes the need to include variations in local living costs in poverty thresholds and assessments (Amendola et al., 2024). This measuring perspective reinforces the article's focus on a price-index channel as a fundamental mechanism connecting climate exposure to poverty in outlying islands.

Ultimately, institutions and public capacity are not only contextual factors in this case. Climate vulnerability is dictated by exposure and adaptive capacity, the latter being affected by infrastructure quality, financial resources and governance. A public finance approach clarifies that intergovernmental transfers and social protection function as mechanisms for geographic insurance and equalization in a climate-impacted context (Musgrave, 1959; Oates, 1972), and their design matters for both efficiency and equity (Bird & Smart, 2002).

2.3. Transportation costs and economic isolation in small island economies

Transport functions as the essential circulatory system for island economies. Increased baseline shipping and logistics expenses diminish competitiveness and elevate consumer prices. Climate change adds a further layer by elevating disruption risk and diminishing infrastructure reliability. Recent research indicates that fluctuations in global shipping costs affect import prices and domestic inflation, suggesting that disruptions in connectivity might diminish purchasing power by increasing delivered prices, in addition to reducing output (Carriere-Swallow et al., 2022).

The literature on SIDS indicates that insularity, remoteness and restricted markets lead to elevated unit transport costs and reduced resilience to external shocks (Briguglio et al., 2009). Models regarding trade expenses and market accessibility illustrate how distance and connectivity influence income levels and specialization trends (S. Redding & Venables, 2004). Climate change significantly amplifies 'distance' via diminishing reliability due to trip cancellations, extended delays, elevated insurance and operational expenses and increased unpredictability in inventory management. In accordance with this, UNCTAD indicates that rerouting and disruptions at maritime chokepoints elevate freight rates, and these increased shipping costs are ultimately transferred to consumers, with SIDS being particularly vulnerable (UNCTAD, 2024).

This demonstrates that an archipelagic NEG expansion constitutes more than only a change of environment. The nature of trade costs transitions from a steadily evolving parameter to an object subject to risk. When trade costs are variable and associated with climate exposure, the correlation between geography and inequality intensifies islands that are both isolated and vulnerable to climate change

experience greater-than-anticipated declines in market access and more significant rises in the cost-of-living during disruptions. An expanding body of literature on port and coastal infrastructure underscores the importance of structured adaptation planning for ports confronting compound hazards, wherein resilience investments specifically aim to enhance the reliability margin critical for island connectivity (Fernandez-Perez et al., 2024). In the Indonesian context, research indicates that Indonesia's maritime freight subsidy modestly and temporarily lowered food prices in remote regions, though the effect dissipated within a year, this shows that connectivity and shipping frictions influence local consumer pricing, even if durable pass-through is limited (Sofiyandi et al., 2023).

2.4. Gaps in literature and study contributions

This article seeks to address several gaps in existing work on island economies, climate risk and spatial inequality. Existing research on island climate vulnerability has largely focused on exposure, impacts and adaptation needs, but has paid less attention to a coherent structural spatial mechanism that links climate shocks to market access, agglomeration, delivered prices and real-income dispersion within a single framework. Recent quantitative spatial climate economics shows that climate change can alter the spatial distribution of activity and welfare through trade and mobility frictions, highlighting the importance of structural spatial models for welfare analysis (Cruz & Rossi-Hansberg, 2024). At the same time, work on spatial price indices and poverty measurement emphasizes that poverty comparisons must be based on purchasing power, not nominal income alone, and calls for better integration of spatial cost-of-living variation into poverty assessments (Amendola et al., 2024). There is still relatively little work that combines these insights to connect climate-induced connectivity disruptions directly to spatial price differentials and poverty risk in small islands.

A second gap concerns how trade costs are modeled in archipelagic and climate-exposed contexts. NEG-inspired research typically treats trade costs as stable or smoothly evolving parameters. In practice, climate risk introduces the possibility of sudden disruptions and heightened variability in connectivity, especially where route redundancy is limited and the loss of a single port can severely restrict access. Evidence on shipping costs and chokepoint disruptions shows that shocks to maritime routes can generate sharp and persistent increases in delivered costs that propagate through supply chains (UNCTAD, 2024). While spatial integrated assessment models (S-IAMs) have begun to explore climate impacts in spatial equilibrium with multiple equilibria and rich dynamics, they often embed complex climate and damage modules that make it difficult to isolate the specific role of delivered-price wedges and spatial cost-of-living differences. There remains room for a simpler, NEG-based structure that treats trade costs explicitly as exposure-sensitive and risk-prone, without requiring a full IAM apparatus.

A third gap relates to island resource and capacity constraints. Much of the island development literature emphasizes limits in land, population size, migration options and human capital, and the role of remittances and external capital flows in shaping agglomeration processes and income dynamics. These factors influence both the strength of agglomeration forces and the capacity to adapt to climate risk, yet they are rarely incorporated into a structural spatial mechanism that also accounts for connectivity disruptions. The model in this article does not attempt to fully address these limitations, but it is designed to be compatible with them: low-market access, small scale and limited redundancy are treated as structural features that make islands more sensitive to climate-induced disruptions in trade costs and agglomeration.

Finally, the policy discussion is often fragmented across instruments. Infrastructure investment, resilience and social protection are frequently analyzed in isolation. Public finance and place-based policy perspectives emphasize the need to view these instruments as a portfolio: place-based connectivity investments aimed at improving long-term productivity and market access; resilience expenditures that reduce the likelihood and impact of disruptions; and social protection that stabilizes consumption and prevents poverty traps during shocks (Balboni, 2025; Kline & Moretti, 2014). There is limited work that embeds this portfolio logic in a structural spatial framework where climate exposure operates through both incomes and local prices, under a binding fiscal envelope.

2.5. What this article adds

This article combines ideas from NEG and spatial climate economics to develop a climate-enhanced NEG framework for archipelagic economies characterized by costly and unreliable connectivity. Within a standard core–periphery structure, climate exposure is introduced as a state variable that affects both effective trade and logistics costs and agglomeration potential. The framework highlights three core theoretical implications: (i) climate exposure increases both the level and volatility of effective inter-island trade and logistics costs, turning market access into a risk-prone object rather than a slowly evolving parameter; (ii) islands with higher exposure experience larger expected losses in market access and agglomeration potential, reducing investment incentives and income potential and (iii) poverty risk may rise through a combined mechanism of lower nominal income and higher local prices, so that welfare losses can be driven or reinforced by cost-of-living effects even when output changes modestly.

Methodologically, the article offers a deliberately stylized, reduced-form complement to richer S-IAMs. It keeps the NEG core simple and transparent, while allowing climate exposure to modify trade costs and agglomeration in a way that can be directly mapped into effective delivered costs, local price indices and real income. This makes it easier to see how exposure-sensitive trade costs, price pass-through and agglomeration elasticities jointly determine welfare outcomes, and under what parameter configurations each channel dominates.

From a policy perspective, the framework embeds a portfolio view of public finance under climate risk. Under a binding fiscal envelope, it shows how connectivity investment, resilience investment and adaptive social protection operate through distinct margins in the spatial equilibrium: connectivity lowers baseline and exposure-sensitive trade friction; resilience slows exposure accumulation and stabilizes agglomeration and social protection stabilizes real incomes when prices spike or earnings fall. The model therefore supports a spatially differentiated policy approach: prioritize connectivity investments where delivered-cost wedges are large and market-access gains are high; prioritize resilience investments where exposure threatens the reliability of ports and transport networks and deploy targeted social protection in remote, exposure-sensitive islands where the cost-of-living channel is particularly strong.

Together, these contributions place the article within the contemporary spatial equilibrium literature on geography and welfare (Allen & Arkolakis, 2023), and the emerging spatial climate literature that emphasizes how shocks interact with trade, mobility and market access (Amendola et al., 2024), while adding a focused treatment of archipelagic connectivity, delivered prices and poverty under climate exposure.

3. Conceptual framework and model implications

This section outlines the mechanisms that underpin the model-based implications developed later in the article. The starting point is conventional NEG, in which spatial outcomes are shaped by market access, transportation costs and agglomeration forces (Fujita et al., 1999; Fujita & Thisse, 2013; Krugman, 1991). In an archipelagic setting, physical climate risk affects these forces by disrupting connectivity, increasing effective trade and logistics costs and weakening the ability of peripheral islands to sustain agglomeration. The welfare consequence is that poverty depends on both income and cost-of-living differences, which is particularly important for small islands where consumption baskets rely heavily on imported goods and substitution possibilities are limited.

3.1. Setup: core, periphery and climate exposure

Consider a core region indexed by zero and peripheral island regions indexed by i. Let C_{it} denote a climate exposure index for region i at time t. In this framework, exposure is a state variable that shifts expected economic conditions, especially those related to connectivity and infrastructure reliability, rather than a measure of realized damage in each period. Consistent with the introduction, C_{it} is interpreted primarily as physical exposure of coastal and transport systems, though it may be correlated with broader transition risks.

Economic outcomes are affected through two spatial channels. First, higher exposure raises both the level and the uncertainty of effective trade and logistics frictions by disrupting port operations, maritime routes and coastal infrastructure. Second, greater exposure weakens agglomeration forces by reducing expected profitability and increasing uncertainty, thereby lowering incentives for firms and workers to locate in

vulnerable areas. Both channels are consistent with the core NEG logic that small changes in market access can generate large differences in equilibrium concentration and welfare through circular causation.

This article represents trade and logistics frictions using the conventional iceberg trade-cost model introduced by Samuelson (1954). The iceberg metaphor captures the idea that a fraction of a shipment is effectively 'melted away' during transportation: if the trade-cost factor between region i and the core 0 is $T_{i0} \geq 1$, then delivering one unit to the destination requires shipping T_{i0} units from the origin. Equivalently, delivered prices satisfy $p_0 = p_i T_{i0}$. This formulation is widely used in trade and economic geography because it provides a tractable way to model trade costs as a multiplicative wedge that directly affects delivered prices and market access (Fujita et al., 1999; Fujita & Thisse, 2013; Krugman, 1991).

In this framework, the iceberg term is interpreted in a generalized way. It encompasses not only distance and fuel, but also reliability-related components, such as port constraints, delays, spoilage and damage, insurance premium and route cancellations. This follows the broader trade-cost literature in treating iceberg costs as a reduced-form representation of multiple frictions rather than literal shipping charges alone (Anderson & van Wincoop, 2004). Real-world transport systems include both fixed and variable cost components and may respond endogenously to economic conditions. Here, rather than modeling a separate transport sector with endogenous pricing, that endogeneity is captured in reduced form by allowing T_{i0} to depend on climate exposure and geography. In an archipelagic context, higher exposure raises effective frictions by interrupting ports and maritime transport and by reducing infrastructure reliability. As a result, greater exposure increases T_{i0}, raises the local price index, and lowers real income.

3.2. Climate-adjusted transportation costs and market access

3.2.1. Climate-adjusted trade and logistics costs

Effective iceberg trade and logistics costs between peripheral region i and the core 0 are modeled as

$$T_{i0t} = \tau_i d_{i0}^{\theta} \exp(\gamma_i C_{it}),$$

where τ_i captures baseline logistics frictions, d_{i0} is distance, θ is the distance elasticity and γ_i measures the sensitivity of trade costs to climate exposure. Differentiating with respect to exposure yields

$$\frac{\partial T_{i0t}}{\partial C_{it}} = \gamma_i T_{i0t}.$$

For $\gamma_i > 0$, this implies a direct monotonic effect: higher exposure increases effective trade and logistics costs proportionally. The archipelagic emphasis is not only that the baseline level $\tau_i d_{i0}^{\theta}$ is high, but also that γ_i is plausibly larger for small islands with limited route redundancy and high dependence on vulnerable coastal infrastructure.

3.2.2. Disruption risk and expected connectivity

Archipelagic transport costs are not only high; they are also unreliable. To incorporate disruption risk explicitly, consider a simple two-state representation of effective trade costs:

$$T_{i0t} = \begin{cases} T_{i0t} & \text{with probability } 1 - p_{it}, \\ \overline{T}_{i0t} > T_{i0t} & \text{with probability } p_{it}, \end{cases}$$

where $p_{it} = p(C_{it})$ is the probability of disruption and $p'(\cdot) > 0$. Under CES preferences, the profitability of serving a market depends on terms of the form $T_{i0t}^{1-\sigma}$, where $\sigma > 1$ is the elasticity of substitution. Expected market access for region i is

$$MA_{it} = \sum_j E\left[T_{ijt}^{1-\sigma}\right] Y_{jt} P_{jt}^{\sigma-1},$$

with Y_{jt} denoting expenditure in destination j and P_{jt} its CES price index. Under the two-state process,

$$E\left[T_{i0t}^{1-\sigma}\right] = (1 - p_{it}) T_{i0t}^{1-\sigma} + p_{it} \overline{T}_{i0t}^{1-\sigma}.$$

Since $\overline{T}_{i0t} > T_{i0t}$ and $1-\sigma < 0$, it follows that $\overline{T}_{i0t}^{1-\sigma} < T_{i0t}^{1-\sigma}$, so

$$\frac{\partial E\left[T_{i0t}^{1-\sigma}\right]}{\partial p_{it}} < 0.$$

With $p_{it} = p(C_{it})$ and $p^{'}(C_{it}) > 0$, the key sign restriction is

$$\frac{\partial MA_{it}}{\partial C_{it}} < 0.$$

Climate exposure therefore reduces expected market access on vulnerable islands, even when baseline distance is held constant. This result is purely theoretical but central: exposure acts as a fragmentation force in the spatial equilibrium.

3.3. Agglomeration under climate risk

NEG emphasizes that agglomeration is sustained by circular causation: better market access raises expected profits, which attracts firms and workers, increases local variety, raises real wages and further strengthens demand (Fujita et al., 1999; Krugman, 1991). When exposure reduces expected market access, similar feedback can operate in reverse. A simple reduced-form representation links profitability to market access:

$$\pi_{it} = \phi(MA_{it}), \phi^{'}(\cdot) > 0.$$

Since

$$\partial MA_{it}/\partial C_{it} < 0,$$

it follows that

$$\frac{\partial \pi_{it}}{\partial C_{it}} < 0.$$

Higher exposure lowers expected profits and thus limits the scale of economic activity that can be sustained on exposed islands. With mobile capital and potentially mobile workers, this contraction reduces local variety and thins markets, contributing to higher local price indices and weaker real wage incentives. The resulting decline in demand reinforces dispersion away from exposed regions, in line with broader geographic economics insights on how trade costs and market access affect the stability of spatial concentration.

3.4. Market fragmentation and regional inequality

In canonical NEG, a reduction in trade costs typically promotes spatial integration and can support convergence between core and periphery (Fujita et al., 1999). In climate-vulnerable archipelagos, exposure works in the opposite direction by increasing effective trade costs and disruption risk. If peripheral islands exhibit higher γ_i, higher baseline frictions τ_i or faster growth in exposure, then expected market access declines more sharply in those locations. This produces greater cross-region dispersion in market access and, consequently, in real incomes.

A simple measure of fragmentation is the cross-region variance of log market access:

$$F_t = \mathrm{Var}_i(\log MA_{it}).$$

When declines in MA_{it} are systematically larger in peripheral islands, F_t rises over time, indicating increasing spatial fragmentation. Because profitability and investment incentives are tightly linked to market access, this process tends to generate divergent income growth paths and higher regional inequality.

3.5. Model implications

The conceptual framework implies three model-based implications. These are theoretical propositions derived from the sign restrictions discussed above, rather than empirical implications already estimated from data. In particular, they rely on the maintained assumptions that $\gamma_i > 0$ and $p'(C_{it}) > 0$, so that climate exposure increases trade frictions and disruption probabilities.

a. **Implication 1 (connectivity and cost-of-living).** Increased climate exposure raises effective trade and logistics costs, which, through higher delivered costs, elevates the local cost-of-living in peripheral islands. The effect is expected to be stronger in more isolated locations and where transport corridors have limited redundancy.
b. **Implication 2 (agglomeration disruption).** Higher exposure reduces expected market access and profitability in peripheral islands, weakening agglomeration economies. This lowers investment incentives and reallocates economic activity toward regions with lower exposure.
c. **Implication 3 (fragmentation and inequality).** By reducing expected market access more strongly in exposed peripheries, climate exposure contributes to market fragmentation and regional inequality, causing real incomes in vulnerable islands to grow more slowly than in less exposed regions.

The next section develops a formal model that makes these mechanisms explicit. Climate exposure enters both effective trade costs and agglomeration potential, and poverty is determined by real-income levels that combine nominal earnings with local prices. The objective is to obtain clear comparative statics that correspond to Implications 1–3 and to clarify the trade-offs in place-based public finance across connectivity investment, resilience expenditure and social protection.

4. Model and comparative statics

This section outlines a cohesive framework that clarifies the underlying mechanisms associated with Implications 1–3. The setup is deliberately stylized yet maintains internal consistency: climate exposure influences (i) effective trade and logistics costs, (ii) agglomeration potential and consequently (iii) output and the local cost-of-living, which collectively determine poverty through real resources. The interplay arises from the progression of climate exposure and the significance of investing in resilience.

Structural note on presentation order. The model is grounded in a standard production environment in which output is the primary object of interest, and trade costs and agglomeration enter as the two key determinants of output and income. They are introduced before the output equation because they are the channels through which climate exposure operates, they must be defined before their role in output and welfare can be stated precisely. This mirrors standard NEG exposition (Fujita et al., 1999; Fujita & Thisse, 2013).

The term 'core region' refers throughout to region 0: the more connected, lower exposure, higher-agglomeration location that anchors inter-island trade. In the Indonesian context, this corresponds conceptually to hub ports, such as Surabaya or Makassar. Peripheral regions ($i = 1, \ldots, N$) are remote islands with higher baseline frictions (τ_i), greater isolation (d_{i0}) and higher exposure sensitivity (κ_i).

On land vs sea distance: d_{i0} in the model refers to 'effective maritime distance', the sea route between region i and the core. This is consistent with the article's focus on inter-island maritime connectivity, where sea transport is the primary determinant of trade costs (Briguglio et al., 2009; UNCTAD, 2024). Intra-island land distance is subsumed in the baseline logistics friction τ_i. Separating land and sea distances would require a more disaggregated spatial structure and is an identified direction for future empirical work.

4.1. Environment, regions and timing

Time is discrete and indexed by $t = 0, 1, 2, \ldots$. The economy consists of a core region indexed by 0 and a set of peripheral island regions indexed by $i \in \{1, \ldots, N\}$. The index 0 refers to the core region and is not a time baseline.

The key state variable is climate exposure in region i, denoted $C_i(t)$. Climate exposure is modeled as a summary index that captures the intensity of climate-related risks relevant to economic activity, particularly those affecting connectivity and infrastructure reliability. Exposure is not interpreted as realized damage in each period. Instead, higher $C_i(t)$ shifts effective trade and logistics costs and weaken the conditions that sustain agglomeration in exposed locations.

To ensure clarity in notation, it is understood that all variables lacking time subscripts are evaluated at the same period t. Cross-region objects, such as distance d_{i0} and baseline logistics parameters, are considered time-invariant throughout the model's horizon unless stated otherwise.

The model is structured to generate comparative statics within a specified timeframe and to facilitate dynamics by accounting for the progression of climate exposure over time. The subsequent subsection presents the motion equation fo $C_i(t)$ and the role of resilience investment.

A deliberate simplifying assumption is that the productivity scale A_i and the technology embedded in Equation (6) are treated as time-invariant. In practice, technology is dynamic: firms learn by doing, workers accumulate skills and infrastructure investment raises total factor productivity. Two arguments justify this treatment. First, the model's objective is to isolate the mechanisms through which climate exposure affects welfare via trade costs and agglomeration, adding endogenous technology would blur these channels without changing the direction of any signed inequality in Sections 4.3–4.9. Second, over the medium-run horizon of the simulation (10–20 periods), treating A_i as approximately stable is a reasonable benchmark for stylized archipelagic environments where baseline productivity growth is slow and market integration is the primary welfare driver.

Introducing learning-by-doing would represent an important extension. If connectivity loss reduces knowledge transfer or specialization opportunities on peripheral islands, the negative welfare effects of climate-induced isolation would be amplified over longer horizons. The static-technology assumption therefore yields a 'conservative estimate' of long-run damage, and future work incorporating dynamic technology evolution is well positioned to test whether persistence and learning externalities matter quantitatively.

4.2. Evolution of climate exposure and resilience investment

The evolution of climate exposure is influenced by the increasing severity of hazards linked to climate change over time. Investing in resilience mitigates future risks by safeguarding essential assets and decreasing vulnerability. The law of motion for exposure in region i is:

$$C_i(t+1) = (1+g_C)\ C_i(t) - \eta_R\ R_i(t), g_C > 0,\ \ \eta_R > 0. \qquad (1)$$

Equation (1) specifies a parsimonious linear law of motion for climate exposure. The term $(1+g_c)C_i(t)$ captures the baseline persistence and trend increase in exposure over time, while $\eta_R R_i(t)$ represents the reduction in future exposure achieved through resilience investment. In this sense, the specification is analogous to a standard stock-adjustment process, in which the state variable evolves according to its own persistence and is partially offset by a policy control. The linear form is adopted deliberately as a reduced form and tractable way to represent exposure dynamics, not as a claim that all underlying climate or infrastructure processes are literally linear.

The parameter g_C captures the exogenous trend in exposure growth. The term $\eta_R R_i(t)$ represents mitigation through resilience investment, where $R_i(t) \geq 0$ includes adaptation spending, such as coastal protection, port hardening, drainage and infrastructure standards, that reduce the vulnerability of transport systems and coastal assets.

To avoid knife-edge or infeasible states, assume exposure is bound below at zero, $C_i(t) \geq 0$. If the right-hand side becomes negative, exposure is interpreted as $C_i(t+1) = 0$. This maintains the economic meaning of the state variable and ensures that resilience investment does not result in negative exposure.

This timing structure elucidates the differences between short- and long-run policy effects. Resilience influences present outcomes only to the extent that it impacts the reliability of current infrastructure or the sensitivity to trade costs, as will be discussed further below. The direct dynamic effect operates through Equation (1) by reducing future exposure, which, in turn, influences the progression of connectivity, agglomeration and welfare over time.

4.3. Climate-adjusted trade and logistics costs

The success of market integration between peripheral islands and the core depends on the costs of trade and logistics. In an archipelagic economy, these costs are indicative of distance, baseline shipping frictions and the reliability of transport systems. The impact of climate exposure on trade and logistics costs is significant, as it leads to the degradation of infrastructure, heightens the risk of disruptions and amplifies uncertainty in maritime connectivity.

Let $T_{i0}(t)$ denote the effective iceberg trade and logistics cost between the peripheral region i and the core region 0. A parsimonious climate-adjusted specification is:

$$T_{i0}(t) = \tau_i \; d_{i0}^{\psi} \exp(\kappa_i \; C_i(t)), \psi > 0, \;\; \kappa_i > 0, \qquad (2)$$

Equation (2) is the model's connectivity bridge: it translates geography and climate exposure into the effective delivered cost of trading with the core. A higher $T_{i0}(t)$ reduces market access for firms and raises delivered prices for households, so this object will appear again in the output equation and the price index. where $\tau_i > 0$ captures baseline logistics frictions in the region i, d_{i0} denotes distance to the core, ψ is the elasticity of trade costs with respect to distance and κ_i governs the sensitivity of effective trade costs to climate exposure. The archipelagic setting motivates allowing κ_i to vary across regions, since route redundancy and the vulnerability of transport corridors differ substantially across islands.

The standard iceberg trade-cost model (Samuelson, 1954) specifies: $T_{i0} = \tau_i \cdot d_{i0}^{\psi}$. This benchmark is the foundation of the NEG literature (Fujita et al., 1999; Krugman, 1991) and models trade costs as a multiplicative delivered-price wedge. However, in archipelagic settings, trade costs are not merely a function of geography, they are also sensitive to infrastructure reliability, which climate exposure actively degrades. Two channels motivate an extension. First, climate-related damage to ports and maritime routes raises the mean level of delivered costs above what distance alone predicts. Second, exposure generates volatility: even when routes are intact, the probability of disruptions rises with exposure, inflating insurance, buffer stock and scheduling costs.

The extension adopted here introduces $C_i(t)$ as an additive shifter in the log of trade costs: $\log T_{i0}(t) = \log \tau_i + \psi \log d_{i0} + \kappa_i C_i(t)$, which is equivalent to Equation (2) in multiplicative form: $T_{i0}(t) = \tau_i \cdot d_{i0}^{\psi} \cdot \exp(\kappa_i C_i(t))$.

The 'exponential functional form' is chosen for three reasons. (i) It preserves the multiplicative iceberg structure: the climate term $\exp(\kappa_i C_i(t))$ scales the baseline cost proportionally, consistent with the idea that exposure amplifies existing frictions. (ii) The log-linear form yields a linear pass-through from exposure to log prices, giving tractable comparative statics. (iii) It preserves $T_{i0}(t) \geq 1$ for all parameter values, ensuring the delivered-price interpretation remains valid.

This formulation 'differs from the standard iceberg model' in two key respects. First, trade costs become time-varying, endogenous to the state variable $C_i(t)$, even when geography is fixed. Second, the sensitivity parameter κ_i is allowed to differ across islands, capturing the empirically relevant fact that small islands with limited route redundancy experience larger delivered-cost responses to the same increase in exposure. In practice, transport costs also include endogenous components (carriers reroute and adjust frequency). However, the reduced-form approach follows the trade-cost literature in treating the iceberg wedge as a comprehensive representation of all frictions, including uncertainty and insurance (Anderson & van Wincoop, 2004). Modeling a full transport sector with endogenous pricing would substantially increase complexity without altering the sign of any key comparative static.

The key comparative static follows immediately:

$$\frac{\partial T_{i0}(t)}{\partial C_i(t)} = \kappa_i \; T_{i0}(t) > 0. \qquad (3)$$

Equation (3) makes the climate mechanism transparent. Exposure increases $T_{i0}(t)$ proportionally, and the proportionality factor κ_i is allowed to differ across islands to reflect different route redundancy and infrastructure vulnerability. In the simulations, islands with larger κ_i experience sharper welfare losses from the same increase in exposure because both incomes and consumer prices become more sensitive to disruption.

Consequently, increased climate exposure elevates effective trade and logistics expenses, thereby diminishing connectivity and market access. The trade-cost channel plays a crucial role in both

production and welfare as it influences (i) the profitability associated with accessing external markets and importing intermediate inputs, and (ii) the final prices of consumption goods on peripheral islands.

If desired, Equation (2) can be interpreted as capturing both the level and the risk component of connectivity in reduced form. In later sections, the implications of higher $T_{i0}(t)$ for prices, output and poverty are made explicit.

4.4. Agglomeration potential under climate exposure

Agglomeration economies encompass the productivity and profitability benefits that emerge when firms and workers congregate. This includes advantages, such as scale economies, more robust input markets, knowledge spillovers and proximity to a concentrated consumer base. In NEG, the interaction of these forces with market access is evident: improved connectivity facilitates clustering, which in turn enhances the local market size (Fujita et al., 1999; Krugman, 1991). In island regions vulnerable to climate change, the unreliability of infrastructure and uncertainty in transportation can diminish these factors by lowering the anticipated benefits of concentration.

Let $\overline{\phi}_i > 0$ denote baseline agglomeration potential in region i, summarizing the strength of clustering forces absent climate exposure. Climate exposure reduces realized agglomeration potential, while resilience investment can offset part of this loss by improving infrastructure reliability. A convenient specification that guarantees positivity is:

$$\phi_i(t) = \overline{\phi}_i \exp(-\rho\ C_i(t)) + \theta_R\ R_i(t), \rho > 0,\ \ \theta_R \geq 0. \qquad (4)$$

The agglomeration potential specified in Equation (4) follows the NEG tradition, in which clustering forces are summarized by an index that depends on market access, local density and profitability. The exponential exposure term captures the idea that repeated disruptions weaken the reliability, supply linkages and market thickness that sustain agglomeration. The resilience term represents a contemporaneous improvement in infrastructure reliability that partially restores these clustering benefits. This specification is a reduced-form way to preserve the core NEG mechanism while keeping the model analytically tractable.

This equation defines agglomeration potential, $\phi_i(t)$, as the model's summary measure of how attractive region i is for clustered activity, firm variety and scale effects. Climate exposure reduces $\phi_i(t)$ through the exponential term, which captures the idea that repeated disruption erodes the effective advantages of clustering by weakening reliability, supply linkages and market thickness. Resilience spending enters additively because it is modeled as a direct improvement to functioning capacity, for example, through port hardening, redundancy and faster recovery, which raises agglomeration potential even when baseline exposure is high. Also, this formula implies that exposure reduces agglomeration potential smoothly and preserves $\phi_i(t) > 0$ for all $C_i(t) \geq 0$ and $R_i(t) \geq 0$. The comparative static with respect to exposure is:

$$\frac{\partial \phi_i(t)}{\partial C_i(t)} = -\rho\ \overline{\phi}_i \exp(-\rho\ C_i(t)) < 0. \qquad (5)$$

Equation (5) clarifies how sensitive agglomeration is to climate exposure. The marginal effect of exposure is negative, and its magnitude depends on both ρ and the current exposure level $C_i(t)$. This makes resilience policy meaningful in two ways: it can slow the growth of exposure over time (via the exposure law of motion earlier) and it can also partially offset current losses in agglomeration capacity through the $\theta_R R_i(t)$ term. The parameter ρ controls how strongly climate exposure erodes the ability of the region i to sustain agglomeration. The term $\theta_R R_i(t)$ captures contemporaneous improvements in reliability from resilience investment. This channel fundamentally differs from the dynamic influence of resilience described by the exposure law of motion in Equation (1). In this context, resilience enhances the economic performance of the current period by reinforcing essential infrastructure and reducing operational unpredictability.

This specification delineates the agglomeration disruption mechanism: climate exposure diminishes clustering forces in affected islands, while resilience can mitigate this impact by safeguarding and stabilizing the infrastructure that underpins concentrated economic activity.

4.5. Output and income

In region *i* output exhibits an increase correlated with agglomeration potential, while it shows a decline in response to effective trade and logistics costs. Agglomeration enhances productivity and profitability by leveraging scale economies and market density, whereas elevated trade costs hinder market access, escalate input expenses and restrict firms' capacity to engage with external markets. A simplified representation that provides clear comparative statics is:

$$Y_i(t) = A_i \ [\phi_i(t)]^{\alpha} \ [T_{i0}(t)]^{-\beta}, A_i > 0, \alpha > 0, \ \ \beta > 0. \ (6)$$

Equation (6) provides a reduced-form income-production block consistent with the core logic of NEG. Output is increasing in agglomeration potential because denser and better-connected locations benefit from thicker input markets, larger demand and scale economies, while it is decreasing in effective trade and logistics costs because higher frictions reduce market access and raise the cost of imported inputs and intermediate goods. The specification is deliberately expressed in terms of nominal income or output, following the standard NEG tradition in which production-side outcomes are defined separately from the local cost-of-living. Real income is then derived later by deflating nominal income with the local price index, which makes it possible to distinguish clearly between the income channel and the cost-of-living channel in the article's welfare analysis.Where A_i captures time-invariant productivity and endowments, α measures the elasticity of output with respect to agglomeration potential and β measures the elasticity of output with respect to effective trade and logistics costs. Equation (6) is the income-production block that connects the earlier objects to measurable outcomes. Higher-agglomeration potential raises output because firms in denser, better-connected places can access larger markets, thicker input supplies and more varieties. Higher effective trade and logistics costs lower output because they reduce market access and raise the cost of inputs and intermediate goods, which is especially relevant in import-dependent islands. This specification matches the model logic in which exposure harms income both by weakening the scale benefits of clustering and by raising the delivered cost of trade.

Equation (6) implies that climate exposure reduces output through two channels: (i) by lowering $\phi_i(t)$ and (ii) by raising $T_{i0}(t)$. Using Equations (3) and (5),

$$\frac{\partial Y_i(t)}{\partial C_i(t)} = Y_i(t)\left[\alpha \ \frac{1}{\phi_i(t)}\frac{\partial \phi_i(t)}{\partial C_i(t)} - \beta \ \frac{1}{T_{i0}(t)}\frac{\partial T_{i0}(t)}{\partial C_i(t)}\right] < 0. \qquad (7)$$

Equation (7) above decomposes the total exposure effect on income into two interpretable components. The first term captures the agglomeration channel: exposure reduces $\phi_i(t)$, lowering output with elasticity α. The second term captures the connectivity channel: exposure raises $T_{i0}(t)$, lowering output with elasticity β. This decomposition is useful for the simulations because it lets the article report whether losses are driven mainly by weakening agglomeration, rising logistics costs or both, and it also shows how policy levers map into income by targeting either $\phi_i(t)$ or $T_{i0}(t)$. Because $\partial \phi_i(t)/\partial C_i(t) < 0$ and $\partial T_{i0}(t)/\partial C_i(t) > 0$ under the maintained parameter restrictions. Thus, higher exposure lowers income potential in peripheral islands by weakening agglomeration and increasing effective trade costs.

Resilience investment can mitigate these effects both contemporaneously (through $\theta_R R_i(t)$ in $\phi_i(t)$) and intertemporally (by reducing $C_i(t+1)$ in the exposure law of motion). Connectivity investment, introduced later as a policy instrument, can be interpreted as reducing baseline frictions embedded in τ_i and therefore lowering $T_{i0}(t)$, raising output through Equation (6).

4.6. Cost-of-living and poverty

In peripheral island regions, connectivity influences welfare by impacting both production and the prices of goods delivered. As trade and logistics costs rise, the delivery costs of consumer goods and intermediate inputs also increase, leading to an uptick in the local price index and an overall rise in the cost-of-living. This channel plays a crucial role in archipelagic environments, particularly where the reliance on imported consumption baskets is significant, and the options for local substitution are restricted.

Let the local cost-of-living (price index) in region i be:

$$P_i(t) = \overline{P}\ [T_{i0}(t)]^{\chi}, \overline{P} > 0, \chi > 0. \qquad (8)$$

Equation (8) provides a reduced-form local price index that is closely related to the transport-cost-inclusive CES price index in Krugman (1991). In the standard NEG setting, iceberg trade costs enter the delivered prices of imported varieties and therefore increase the local cost-of-living. The specification used here compresses that mechanism into a tractable pass-through form, in which the parameter κ summarizes the elasticity of the local price index with respect to effective trade and logistics costs. The purpose of this reduced-form mapping is to preserve the core Krugman intuition that higher trade frictions raise local prices, while keeping the model simple enough to isolate the welfare consequences of climate-driven connectivity disruptions in island regions. This formulation also clarifies why welfare must be evaluated in real rather than nominal terms, since exposure affects households not only through income but also through the local cost-of-living.

Furthermore, the price index is the second core outcome in the model because it converts nominal income into purchasing power. In an archipelagic setting, the price index is a spatial cost-of-living index that responds sharply to connectivity. When climate exposure increases effective trade costs or reduces shipping reliability, delivered prices of imported varieties rise, and variety availability can shrink, both of which increase the local price index. This mechanism is central for poverty analysis because it implies that exposure can raise poverty risk even when nominal incomes do not fall proportionally, particularly in remote islands with limited substitution options and high import dependence, where χ captures the pass-through from effective trade costs into local prices. From Equation (3), it follows that

$$\frac{\partial P_i(t)}{\partial C_i(t)} = \chi\ \overline{P}\ \left[T_{i0}(t)\right]^{\chi-1} \frac{\partial T_{i0}(t)}{\partial C_i(t)} > 0. (9)$$

Real income is defined as nominal income deflated by the cost-of-living:

$$RI_i(t) \equiv \frac{Y_i(t)}{P_i(t)}. \qquad (10)$$

Using Equations (6) and (8), real income can be written as:

$$RI_i(t) = \frac{A_i}{\overline{P}}\ \left[\phi_i(t)\right]^{\alpha} \left[T_{i0}(t)\right]^{-(\beta+\chi)}. (11)$$

The equation above makes the joint income and price mechanism explicit: higher trade costs reduce real income both by lowering output (β) and by raising the cost-of-living (χ). Taking logs and differentiating with respect to exposure yields a single comparative static that anchors the poverty mechanism:

$$\frac{\partial \log RI_i(t)}{\partial C_i(t)} = \alpha\ \frac{1}{\phi_i(t)} \frac{\partial \phi_i(t)}{\partial C_i(t)} - (\beta+\chi)\ \frac{1}{T_{i0}(t)} \frac{\partial T_{i0}(t)}{\partial C_i(t)} < 0, \qquad (12)$$

since $\partial \phi_i(t)/\partial C_i(t) < 0$ and $\partial T_{i0}(t)/\partial C_i(t) > 0$. This inequality formalizes the key claim: climate exposure reduces real income through both lower income potential and higher local prices.

Poverty is modeled as a decreasing function of real income:

$$\Pi_i(t) = \Pi(RI_i(t)),\ \ \Pi'(\cdot) < 0, \qquad (13)$$

so that $\partial \Pi_i(t)/\partial C_i(t) > 0$ under the conditions above.

If social protection is introduced as an instrument that augments disposable resources, it can be incorporated directly by defining disposable income as $\tilde{Y}_i(t) = Y_i(t) + \sigma S_i(t)$ with $\sigma > 0$, and redefining real income as $\tilde{Y}_i(t)/P_i(t)$. This extension clarifies how transfers can reduce poverty even when exposure raises prices.

4.7. Fragmentation and inequality

The effects of climate exposure differ by region, influenced by variations in baseline isolation and climate sensitivity, which exhibit spatial inconsistencies. The model demonstrates that there are diverse

shifts in real income among various regions, which could result in heightened market fragmentation and regional disparities. In an archipelagic economy, this mechanism is especially viable because of the differing baseline logistics frictions present across various islands τ_i, a greater distance from the core d_{i0}, and higher exposure sensitivity κ_i. The observed variations indicate that a uniform increase in exposure could result in considerably larger effective trade-cost escalations and more pronounced price pass-through in the most isolated and vulnerable islands.

A parsimonious inequality object is the cross-region dispersion of log real income:

$$\mathcal{I}(t) \equiv \mathrm{Var}_i(\log RI_i(t)), \qquad (14)$$

where $RI_i(t) = Y_i(t)/P_i(t)$ is defined in Equation (10). Using Equation (11), log real income can be written as:

$$\log RI_i(t) = \log\left(\frac{A_i}{\overline{P}}\right) + \alpha \log \phi_i(t) - (\beta + \chi)\log T_{i0}(t). \qquad (15)$$

Because $T_{i0}(t) = \tau_i d_{i0}^{\psi} \exp(\kappa_i C_i(t))$, regions with higher τ_i, greater d_{i0} or higher κ_i experience larger declines in real income for a given increase in exposure. Likewise, if exposure $C_i(t)$ rises faster in peripheral islands over time, the negative effect on real income is amplified in those locations.

Market fragmentation in this framework can be interpreted as increasing dispersion in effective connectivity or real income. Since climate exposure changes the connectivity term $\log T_{i0}(t)$ heterogeneously across space, Equation (15) implies that $\mathcal{I}(t)$ will rise when exposure growth or exposure sensitivity is systematically larger in peripheral islands. In this context, climate change leads to core–periphery divergence not merely by diminishing average welfare, but also by expanding the spatial distribution of welfare outcomes.

This effect is directly related to Implication 3. When exposure is spatially uneven and interacts with distance and baseline logistics frictions, climate risk can exacerbate regional inequality through a combination of (i) larger effective trade-cost penalties, (ii) greater cost-of-living increases and (iii) diminished agglomeration potential in vulnerable islands.

4.8. Policy instruments and the public finance constraint

The model is structured to address a specific public finance problem in a geographically decentralized setting. Policy operates through three instruments: connectivity investment, resilience investment and social protection. These instruments affect welfare through different but related channels, and their use is limited by fiscal capacity.

4.8.1. Connectivity investment

Connectivity investment $I_i(t)$ includes infrastructure and logistics improvements that reduce baseline trade and shipping frictions, such as port upgrades, coastal transport reliability and measures that increase route redundancy. In the model, this instrument operates by reducing baseline logistics frictions, as shown in Equation (2). A reduced-form representation is:

$$\tau_i = \tau(I_i(t)), \tau'(\cdot) < 0. \qquad (16)$$

A broader interpretation allows connectivity investment to reduce the climate sensitivity of trade costs by improving reliability and redundancy:

$$\kappa_i = \kappa(I_i(t)), \kappa'(\cdot) \leq 0. \qquad (17)$$

Under either representation, higher $I_i(t)$ reduces effective trade and logistics costs $T_{i0}(t)$, lowers the cost-of-living $P_i(t)$, and raises output $Y_i(t)$ through improved market access.

4.8.2. Resilience investment

Resilience investment $R_i(t)$ enters in two ways. First, it reduces future exposure through the law of motion in Equation (1). Second, it can improve contemporaneous agglomeration potential by stabilizing

infrastructure reliability in Equation (4). This structure captures the idea that resilience provides both a short-run reliability benefit and a long-run mitigation benefit.

4.8.3. Social protection

Social protection, $S_i(t)$, is interpreted as a policy instrument that stabilizes welfare when exposure raises prices or disrupts incomes. As noted in Section 4.6, it can be incorporated as an augmentation of disposable resources:

$$\tilde{Y}_i(t) = Y_i(t) + \sigma S_i(t), \sigma > 0, \qquad (18)$$

so that the relevant welfare object becomes $\tilde{Y}_i(t)/P_i(t)$. This formulation clarifies why adaptive transfers can be especially valuable in remote and exposed regions where the cost-of-living channel is strong.

4.8.4. Fiscal envelope and budget constraint

The three instruments are constrained by an intergovernmental and local fiscal budget:

$$I_i(t) + R_i(t) + S_i(t) \leq B_i(t), \qquad (19)$$

where $B_i(t)$ represents the total fiscal envelope available to region i, financed through own-source revenues and intergovernmental transfers. This constraint highlights the core tradeoff in public finance. Connectivity and resilience investments primarily influence long-run income potential and the evolution of exposure, while social protection primarily stabilizes welfare in the short run. In fiscally constrained island peripheries, the composition of public spending becomes central to poverty outcomes, not just the level of spending.

The next subsection summarizes the comparative statics implied by the model and links them back to the implications stated in Section 3.

4.9. Summary of comparative statics and links to implications

The model yields a set of transparent comparative statics that map directly to the implications.

1. **Exposure increases effective trade costs and the cost-of-living.**

From Equations (2) and (3), $\partial T_{i0}(t)/\partial C_i(t) > 0$. Since the cost-of-living depends positively on trade costs in Equation (8), it follows that $\partial P_i(t)/\partial C_i(t) > 0$. This establishes the connectivity–price channel emphasized in Implication 1.

2. **Exposure weakens agglomeration and lowers output.**

From Equations (4) and (5), $\partial \phi_i(t)/\partial C_i(t) < 0$. Output depends positively on agglomeration potential and negatively on trade costs in Equation (6), implying $\partial Y_i(t)/\partial C_i(t) < 0$ as shown in Equation (7). This formalizes the agglomeration disruption and income channel emphasized in Implication 2.

3. **Exposure reduces real income and increases poverty.**

Real income is given by Equation (11). The key object is the log-derivative in Equation (12), which shows that exposure lowers real income through both channels: it lowers $\phi_i(t)$ and raises $T_{i0}(t)$, with the latter also raising the cost-of-living. Since poverty is decreasing in real income in Equation (13), it follows that exposure increases poverty under the maintained parameter restrictions. This provides a unified mathematical statement for the joint income and cost-of-living poverty mechanism.

4. **Heterogeneous exposure and sensitivity generate fragmentation and inequality.**

Equation (15) shows that log real income depends negatively on $\log T_{i0}(t)$, which contains τ_i, d_{i0}, and $\kappa_i C_i(t)$. If exposure growth, baseline frictions or exposure sensitivity are systematically larger in

peripheral islands, then the decline in real income is larger there, implying rising dispersion in real income $\mathcal{I}(t)$ as defined in Equation (14). This corresponds to market fragmentation and increasing regional inequality as stated in Implication 3.

5. **Policy instruments operate through distinct margins under a fiscal constraint.**

Connectivity investment lowers baseline frictions τ_i (and potentially κ_i), resilience investment reduces future exposure and can improve contemporaneous reliability and social protection stabilizes disposable resources when prices spike, or incomes fall. These instruments are jointly constrained by the budget in Equation (19), making the policy problem inherently one of fiscal allocation and place-based targeting.

The implications outlined create a clear connection between climate exposure and its effects on connectivity disruptions, agglomeration outcomes, real income, poverty and inequality in archipelagic economies. Additionally, they highlight the need for a diverse range of place-based policy instruments, particularly in contexts with constrained fiscal capacity.

5. Policy implications for place-based poverty reduction under climate risk

Climate exposure increases poverty in peripheral island regions through two interconnected mechanisms in the model: it diminishes income potential by reducing market access and undermining agglomeration, while simultaneously elevating the cost-of-living by raising effective trade and logistics expenses.

The assertion is that assessing policy effectiveness should prioritize actual results over merely nominal income, as price influences can be significant in isolated islands where delivery expenses significantly impact the local price index.

5.1. A Policy portfolio: connectivity, resilience and social protection

The model naturally motivates a portfolio of three instruments: connectivity investment $I_i(t)$ that lowers baseline logistics frictions and, where relevant, reduces climate sensitivity; resilience investment $R_i(t)$ that reduces exposure over time and stabilizes economic activity and social protection $S_i(t)$ that stabilizes household resources when disruptions increase prices and reduce incomes.

In a decentralized context, the portfolio is limited by a fiscal envelope supported by own-source revenues and intergovernmental transfers. Consequently, the central policy question revolves around the allocation of a constrained budget among long-term productivity, risk mitigation and short-term stabilization efforts.

5.1.1. Connectivity investment

Connectivity investment $I_i(t)$ refers to public spending that lowers baseline isolation and improves the reliability of inter-island logistics, including port upgrades, shipping facilities, coastal transport connections and measures that increase route redundancy. In the model, the direct role of connectivity investment is to reduce baseline logistics frictions embedded in the effective trade-cost term. Formally, baseline frictions are allowed to depend on investment:

$$\tau_i = \tau\ (I_i(t)), \tau^{'}(\cdot) < 0. \qquad (20)$$

Therefore, higher $I_i(t)$ lowers τ_i and therefore lowers effective trade and logistics costs $T_{i0}(t)$ in Equation (2). This has two immediate welfare implications. First, because the local price index is increasing in trade costs, $P_i(t) = \overline{P}[T_{i0}(t)]^{\chi}$, reductions in τ_i reduce the cost-of-living and dampen price pass-through in peripheral islands. Second, because output is decreasing in trade costs in Equation (6), reductions in τ_i raise income potential by improving market access and lowering delivered input costs.

Connectivity investment can also reduce the climate sensitivity of trade costs by increasing redundancy and reliability in transport systems. This is captured by allowing:

$$\kappa_i = \kappa(I_i(t)), \kappa^{'}(\cdot) \leq 0, \qquad (21)$$

so that higher $I_i(t)$ reduces the degree to which exposure translates into higher effective trade costs. This channel is particularly important in archipelagic settings where disruption risk is a central

component of the climate mechanism: when redundancy is low, climate exposure translates into larger effective increases in delivered costs. In the model, lowering κ_i reduces the local price index's exposure elasticity and increases the stability of market access over time.

The real-income object in Equation (11) clarifies why connectivity investment can be a poverty-reduction instrument rather than only a growth instrument:

$$RI_i(t) = \frac{A_i}{\overline{P}} \left[\phi_i(t)\right]^{\alpha} \left[T_{i0}(t)\right]^{-(\beta+\chi)}.$$

Reducing $T_{i0}(t)$ raises real income through both the income channel (β) and the cost-of-living channel (χ). This is especially important for remote islands where the price index is highly sensitive to delivered costs and where households face large welfare losses from even temporary connectivity disruptions.

The model therefore predicts that the returns to connectivity investment are highest in regions where baseline trade frictions and distance are large (high τ_i and d_{i0}), where price pass-through is strong (high χ), and where exposure sensitivity is high (high κ_i). In such locations, connectivity investments not only enhance long-term income potential but also reduce the vulnerability of real income to climate-driven disruptions.

5.1.2. Resilience investment

Resilience investment $R_i(t)$ refers to public spending that reduces vulnerability of critical assets and improves the reliability of infrastructure under climate stress, including coastal protection, port hardening, drainage and adaptation standards for transport and public facilities. In the model, resilience investment affects welfare through two distinct margins: an intertemporal exposure-reduction channel and a contemporaneous reliability channel.

First, resilience reduces future climate exposure through the law of motion introduced in Section 4:

$$C_i(t+1) = (1+g_C)C_i(t) - \eta_R R_i(t), \;\; g_C > 0, \quad \eta_R > 0. \qquad (1 \text{ revisited})$$

Since effective trade and logistics costs rise with exposure in Equation (2), and agglomeration potential falls with exposure in Equation (4), lowering future $C_i(t)$ slows the growth of trade-cost penalties and mitigates the erosion of agglomeration potential. This is the long-run mitigation logic: resilience investment raises future real income by preventing exposure from compounding.

Second, resilience can improve contemporaneous economic functioning by stabilizing infrastructure reliability and reducing operational uncertainty. This is captured in the agglomeration potential term:

$$\phi_i(t) = \overline{\phi}_i \exp\left(-\rho C_i(t)\right) + \theta_R R_i(t), \rho > 0, \;\; \theta_R \geq 0, \qquad (4 \text{ revisited})$$

where $\theta_R R_i(t)$ represents the current-period reliability benefit from resilience spending. This term is intentionally separated from the dynamic channel in Equation (1). The dynamic channel reduces future exposure, while the contemporaneous channel improves current economic conditions by preserving the functionality of ports, roads and related systems that support dense economic activity and market depth. Through Equation (6), higher $\phi_i(t)$ directly increases output, and through Equation (11), it raises real income and lowers poverty.

Real-income representation is again useful for interpretation. From

$$RI_i(t) = \frac{A_i}{\overline{P}} \left[\phi_i(t)\right]^{\alpha} \left[T_{i0}(t)\right]^{-(\beta+\chi)},$$

resilience raises real income by increasing $\phi_i(t)$ today and by reducing future $C_i(t)$, thereby moderating both the agglomeration channel and the trade-cost channel. This makes resilience particularly important in high-exposure regions where $C_i(t)$ is already elevated or growing rapidly. In those locations, delaying resilience investment allows exposure to compound, making later interventions more expensive to achieve the same reduction in effective trade costs and vulnerability.

The model indicates that resilience spending yields the greatest marginal returns in contexts of high exposure, where the sensitivity of trade costs to exposure is significant (with κ_i is large), and where the economic system relies substantially on a limited number of vulnerable assets, such as a few ports or coastal corridors. In these contexts, resilience serves as more than just a policy for adaptation. This policy

is grounded in a place-based development framework, as it safeguards market access, maintains agglomeration potential and ensures the stabilization of real incomes over time.

5.1.3. Social protection and adaptive transfers

Social protection $S_i(t)$ is introduced as a stabilization instrument that protects welfare when climate exposure increases the cost-of-living or disrupts incomes. In the framework, the poverty mechanism operates through real income $RI_i(t) = Y_i(t)/P_i(t)$. This implies that even temporary increases in the local price index $P_i(t)$ can raise poverty if households cannot smooth consumption. This channel is especially relevant for remote islands where delivered prices are highly sensitive to trade and logistics costs and where short-run substitution options are limited.

To incorporate social protection directly, define disposable resources as nominal income plus a transfer component:

$$\tilde{Y}_i(t) = Y_i(t) + \sigma S_i(t), \sigma > 0, \qquad (18 \text{ revisited})$$

so that the relevant welfare object becomes $\tilde{Y}_i(t)/P_i(t)$. Under this formulation, transfers can reduce poverty even when exposure raises prices, and the effectiveness of transfers depends on the strength of the cost-of-living channel. When χ is large in the price equation $P_i(t) = \overline{P}[T_{i0}(t)]^{\chi}$, price movements contribute strongly to changes in real income, increasing the value of stabilization.

This representation also highlights a design implication. If exposure primarily affects welfare through increases in delivered prices, then transfer programs that are insensitive to local price dynamics may be less effective at stabilizing real consumption. In highly exposed and remote regions, adaptive transfers, such as temporary top-ups triggered by disruption indicators or indexation to local price changes, can provide stronger protection than fixed nominal transfers. In the context of the model, an 'adaptive' design can be interpreted as making $S_i(t)$ respond to shocks in $P_i(t)$ or to the realizations of disruption risk embedded in $T_{i0}(t)$.

Social protection is distinct from connectivity and resilience in its fundamental aspect. The interplay between connectivity and resilience influences trade costs and exposure, thereby altering the long-term trajectory of real income. Social protection primarily reallocates current disposable resources, thereby stabilizing welfare during periods of disruption. In a decentralized fiscal environment, a critical tradeoff emerges: regions that require stabilization the most typically possess the least capacity for self-financing. This dynamic enhances the importance of intergovernmental transfers as a form of spatial insurance.

Collectively, these factors support the notion of considering social protection as a crucial component of a policy portfolio. In remote and exposed islands, stabilization policies can effectively mitigate temporary price increases and income losses, preventing them from evolving into lasting poverty, out-migration and further market thinning. This intervention is crucial in counteracting the fragmentation mechanism highlighted in the model.

5.2. The fiscal allocation problem in a decentralized system

The three instruments described above operate through different margins, but they are jointly constrained by fiscal capacity. Let the available fiscal envelope in region i at time t be $B_i(t)$, financed through own-source revenues and intergovernmental transfers. The government's allocation problem is captured by the budget constraint:

$$I_i(t) + R_i(t) + S_i(t) \leq B_i(t). \qquad (19 \text{ revisited})$$

This constraint makes the policy problem explicitly public finance. Connectivity investment $I_i(t)$ primarily lowers baseline trade frictions τ_i and, where relevant, reduces climate sensitivity κ_i, affecting both the level and volatility of effective trade costs. Resilience investment $R_i(t)$ reduces future exposure through the law of motion for $C_i(t)$ and can improve contemporaneous reliability through $\phi_i(t)$. Social protection $S_i(t)$ stabilizes disposable resources in the event of price spikes or income disruptions. Since these uses of funds compete within the same envelope, the composition of spending becomes central for poverty outcomes, not only the level of spending.

A convenient way to express the allocation logic is to compare the marginal poverty reduction per unit of spending across instruments, given local conditions. Since poverty is decreasing in real income, $\Pi_i(t) = \Pi(RI_i(t))$ with $\Pi'(\cdot) < 0$, the marginal value of an instrument can be interpreted through its effect on real income $RI_i(t)$. In the framework, regions differ in parameters that determine these marginal effects, including baseline isolation τ_i, distance d_{i0}, climate sensitivity κ_i, exposure $C_i(t)$ and price pass-through χ. The effectiveness of a marginal dollar is influenced by these parameters, which assess whether it is better allocated toward lowering trade costs (connectivity), enhancing future resilience or stabilizing disposable resources (social protection).

This interpretation leads to a straightforward targeting guideline: direct the next funds toward the instrument that generates the greatest increase in real income (or the most significant reduction in poverty) per unit of expenditure, considering the local parameter setup. In remote islands characterized by significant price pass-through, implementing policies aimed at reducing delivered costs can yield substantial benefits. This is due to their ability to enhance real income via both the output channel and the cost-of-living channel as outlined in Equation (11). In areas characterized by significant exposure growth or heightened vulnerability, resilience plays a crucial role by mitigating the compounding of exposure. This, in turn, lessens future trade-cost penalties and safeguards the potential for agglomeration. In areas where disruptions lead to significant short-term welfare losses, adaptive social protection may be more effective as it directly stabilizes real consumption in response to increases in $P_i(t)$.

This allocation issue is influenced by considerations of equity and fiscal capacity. Areas characterized by significant exposure and elevated baseline isolation often experience constraints in their own-source revenue and encounter increased unit costs associated with the provision of infrastructure and services. A decentralized system establishes a basis for intergovernmental transfers that serve as mechanisms for place-based equalization and spatial insurance. This enables regions that are vulnerable and financially limited to fund the necessary policy mix aimed at averting climate-induced disparities in welfare outcomes.

5.3. Fragmentation, inequality, and the case for a policy portfolio

The model suggests that climate exposure can exacerbate regional inequality by widening the dispersion of real income across regions. In Section 4, inequality was represented as dispersion in log real income:

$$\mathcal{I}(t) = \mathrm{Var}_i(\log RI_i(t)),$$

with $\log RI_i(t) = \log(A_i/\overline{P}) + \alpha \log \phi_i(t) - (\beta + \chi)\log T_{i0}(t)$. Since $T_{i0}(t)$ embeds baseline frictions τ_i, distance d_{i0} and the exposure term $\kappa_i C_i(t)$, heterogeneous exposure and heterogeneous sensitivity imply heterogeneous declines in real income. In particular, if peripheral islands have higher baseline frictions, higher exposure or higher sensitivity of trade costs to exposure, then $\mathcal{I}(t)$ increases as climate risk intensifies.

The policy portfolio described above can be interpreted as an inequality-mitigating strategy because each instrument reduces dispersion in the key terms that drive real income. Connectivity investment reduces baseline frictions τ_i and can reduce sensitivity κ_i, lowering the effective 'distance penalty' that remote regions face in both normal times and disruption periods. Resilience investment reduces future exposure $C_i(t)$ and stabilizes agglomeration potential $\phi_i(t)$, limiting the degree to which vulnerable regions fall behind over time. Social protection stabilizes disposable resources when disruptions temporarily raise prices or reduce incomes, preventing short-run shocks from generating persistent poverty and out-migration that would further thin local markets and reinforce fragmentation.

This interpretation underscores the necessity of assessing climate policy in island peripheries by focusing on actual outcomes and spatial distribution, rather than merely considering average effects. In scenarios where the cost-of-living channel is robust, implementing policies aimed at lowering delivered costs can effectively mitigate inequality. This occurs even in the absence of significant increases in nominal output, as such policies directly enhance real incomes in the most marginalized areas. In a similar vein, investing in resilience can yield significant distributional benefits by mitigating the risk of compounding exposure in areas characterized by the highest vulnerability and the least fiscal capacity.

The model indicates that climate change can lead to core–periphery divergence in archipelagic economies through a systematic mechanism: increased exposure elevates effective trade costs and diminishes agglomeration more significantly in vulnerable islands, resulting in greater dispersion of real

income. A portfolio that integrates connectivity, resilience and adaptive social protection has the potential to mitigate this divergence by lowering the exposure penalty, reducing price pass-through and stabilizing welfare during periods of disruption.

5.4. Empirical implications and a path to measurement

The framework, while theoretical in nature, produces testable implications that correspond to observable entities and outlines a practical agenda for forthcoming empirical research. The primary implication posits that climate exposure influences poverty via both production mechanisms and the cost-of-living channel, with the efficacy of this channel being contingent upon baseline isolation and transport sensitivity.

Initially, it is important to establish that climate exposure correlates with diminished connectivity. In an archipelagic context, this can be assessed through various proxies, including port throughput, shipping frequency, freight delays, route cancellations or indicators of infrastructure disruption. In the model, exposure increases effective trade and logistics costs $T_{i0}(t)$; empirically, this implies that exposed islands should experience larger increases in transport frictions and larger volatility in connectivity indicators.

Second, climate exposure should raise local prices and the cost-of-living. The price-index relationship $P_i(t) = \overline{P}[T_{i0}(t)]^{\chi}$ implies that increases in effective trade costs should pass-through into higher consumer prices, particularly for traded and import-dependent goods. This implies that islands with higher exposure should experience larger increases in price levels and larger price spikes during disruption episodes, holding income constant.

Third, poverty should respond to climate exposure through both income and price effects. The key prediction is that changes in real income $RI_i(t) = Y_i(t)/P_i(t)$ drive poverty. Empirically, this implies that poverty should rise not only when exposure depresses income growth but also when exposure raises local prices. The magnitude of the poverty response should be larger in regions with stronger price pass-through (high χ) and higher baseline isolation (high τ_i and d_{i0}).

Fourth, resilience spending is expected to reduce the extent of pass-through. Resilience diminishes exposure over time and enhances current reliability. Therefore, areas with greater investment in resilience should exhibit weaker correlations between exposure and (i) connectivity disruptions, (ii) local price spikes and (iii) poverty responses. This serves as a direct empirical counterpart to the mitigation and reliability pathways outlined in Equations (1) and (4).

The model suggests variability in the effectiveness of policy instruments. Investments in connectivity are likely to yield more significant real-income increases in remote islands where there is a robust price pass-through mechanism. Resilience is of paramount importance in contexts where exposure is significant or rapidly increasing. Social protection is crucial in contexts where disruptions lead to significant immediate price hikes or income reductions. This approach encourages interaction designs that examine if policy effects differ consistently based on factors, such as isolation, exposure and price sensitivity.

The implications indicate a systematic approach to measurement through the utilization of subnational panels. Exposure measures can be developed from indices of hazard and vulnerability; connectivity can be represented by transport and port indicators; prices can be assessed through regional CPI or commodity price series; poverty outcomes can be evaluated using official poverty rates and spending categories can be analyzed using government budgets broken down into infrastructure, resilience-related spending and social assistance. An effective empirical design would subsequently examine if the exposure–poverty relationship is more pronounced in areas where the model indicates it should be most significant, and whether investments in resilience and connectivity mitigate the transmission from exposure to prices and poverty.

6. Calibration and quantitative illustration

6.1. Purpose and scope of the quantitative illustration

This section provides a quantitative illustration designed to make the mechanisms developed in Section 4 more transparent and interpretable. The article remains deliberately theory-driven, and all numerical results are generated from the theoretical model rather than estimated from data.

The quantitative exercise is best interpreted as a disciplined calibration. Parameter values are chosen to be economically plausible, consistent with the model's internal restrictions, and broadly aligned with ranges reported in related spatial and trade literatures, while preserving a clear core–periphery asymmetry. The objective is to translate the comparative statics of the framework into magnitudes and trajectories, showing how an increase in climate exposure propagates through effective trade and logistics costs, agglomeration incentives, the local price index and real income, and how these changes map into poverty within the model. The illustration is therefore presented as a model-based prediction exercise rather than as an empirical test.

Indonesia provides the motivating institutional and geographic context for the exercise, but it is not the direct object of structural estimation. The two simulated locations, labeled core and periphery, are not specific Indonesian provinces. Instead, they represent a stylized archipelagic setting in which a remote island faces higher baseline logistics frictions and greater exposure sensitivity than a core region. Indonesia is used as an empirically motivated illustrated setting because it combines the key features emphasized by the model: archipelagic geography, strong dependence on inter-island maritime connectivity, climate exposure concentrated in coastal infrastructure and a decentralized fiscal system with resource-constrained outer islands.

Accordingly, the parameter values reported in Table 1 'are not estimated from Indonesian administrative, price, climate, or household data'. They are chosen to generate transparent model dynamics under plausible assumptions and to illustrate the channels through which climate exposure affects delivered prices, market access, agglomeration and poverty. In this sense, the exercise is intended to be generic to archipelagic or small-island environments with similar structural characteristics, rather than country-specific.

The model is most directly applicable to archipelagic economies because the disruption mechanism is motivated by inter-island maritime dependence. Its use beyond such settings would require adaptation. In landlocked or well-connected continental economies, route redundancy is typically greater and exposure-sensitive transport frictions may be weaker, which would reduce the strength of both the price and agglomeration channels. The framework could nonetheless be extended to other contexts in which climate-related infrastructure disruption raises trade costs, such as corridor-dependent inland economies, but that application would require re-parameterizing the trade-cost and exposure-sensitivity terms for the relevant infrastructure environment.

Table 1. Baseline parameterization for quantitative illustration.

Panel	Parameter	Symbol	Value	Notes
Elasticities	Agglomeration weight	α	0.30	Exponent on agglomeration potential in $Y_i(t)$
Elasticities	Trade-cost elasticity of output	β	0.80	Exponent on effective trade costs in $Y_i(t)$
Elasticities	Price pass-through elasticity	χ	0.60	Pass-through from $T_{i0}(t)$ to $P_i(t)$
Elasticities	Distance elasticity in trade costs	ψ	1.00	Exponent on distance term in $T_{i0}(t)$
Elasticities	Exposure sensitivity of agglomeration	ρ	0.20	Controls how exposure reduces $\phi_i(t)$
Exposure dynamics	Initial exposure	$C(0)$	0.20	Initial exposure level
Exposure dynamics	Exposure growth parameter	g_C	0.05	Baseline exposure growth in law of motion
Exposure dynamics	Resilience intensity (periphery counterfactual)	R_1	0.60	Applied only to periphery counterfactual
Exposure dynamics	Resilience effectiveness	η_R	0.03	Reduces exposure growth in law of motion
Exposure dynamics	Direct resilience term in agglomeration (set to zero)	θ_R	0.00	Set to zero in this illustration
Exposure dynamics	Price normalization	$\bar{P}$	1.00	Normalization for $P_i(t)$
Core ($i = 0$)	Baseline friction	τ_0	1.05	Baseline effective trade friction
Core ($i = 0$)	Distance to core market	d_{00}	1.00	Distance term entering $T_{00}(t)$
Core ($i = 0$)	Exposure sensitivity of trade costs	κ_0	0.05	Sensitivity in $\exp(\kappa_i C_i(t))$
Core ($i = 0$)	Baseline agglomeration potential	$\bar{\phi}_0$	1.10	Scale term in $\phi_0(t)$
Core ($i = 0$)	Productivity scale	A_0	1.00	Scale term in $Y_0(t)$
Periphery ($i = 1$)	Baseline friction	τ_1	1.25	Higher baseline friction than core
Periphery ($i = 1$)	Distance to core market	d_{10}	2.00	Higher isolation than core
Periphery ($i = 1$)	Exposure sensitivity of trade costs	κ_1	0.15	Higher exposure sensitivity than core
Periphery ($i = 1$)	Baseline agglomeration potential	$\bar{\phi}_1$	1.00	Scale term in $\phi_1(t)$
Periphery ($i = 1$)	Productivity scale	A_1	0.90	Lower productivity than core

Note: Parameters are selected for transparency and to preserve a core–periphery asymmetry. The illustration is not an estimated calibration for Indonesia.

6.2. Objects simulated and mapping to the model

The quantitative illustration simulates a restricted set of objects that map one-to-one to the theoretical structure in Section 4. The selection is deliberate: each simulated series corresponds to a specific mechanism in the framework, and together they span the full transmission chain from climate exposure to welfare and poverty within the model.

The first object is the model's measure of 'effective trade and logistics costs' between a location i and the core market (denoted $T_{i0}(t)$). This object is the formal representation of connectivity conditions. In the framework, climate exposure enters this margin through the exposure sensitivity parameter, so that higher exposure increases effective trade costs and weakens market access. Since the analysis is comparative static in exposure, this object is the natural starting point for the simulation: it is the wedge that converts climate risk into an economic distance penalty.

The second object is the 'local price index' $P_i(t)$, which maps trade and logistics costs into the cost-of-living through the price pass-through elasticity. This mapping is essential for the article's central claim that welfare losses in peripheral islands may be amplified by prices, not only by income. In log form, the price-index response is proportional to the change in effective trade costs, with the proportionality governed by the pass-through parameter. The simulation therefore treats $P_i(t)$ as the direct empirical analogue of the cost-of-living channel that policy discussions often omit.

The third object is 'agglomeration potential' $\phi_i(t)$. This object captures the productivity environment that supports clustering, investment and scale economies. Climate exposure reduces agglomeration potential in the model by lowering reliability and increasing expected disruption, which weakens the incentive for firms and workers to locate in exposed places. In the simulation, $\phi_i(t)$ is computed as an exposure-sensitive index, and it is allowed to respond to resilience policy in the counterfactual through the same structure specified in the theoretical model.

The fourth object is 'nominal income (or output potential)' $Y_i(t)$, which combines the agglomeration channel and the market-access channel. In the framework, $Y_i(t)$ is increasing in agglomeration potential and decreasing in effective trade costs. Simulating $Y_i(t)$ makes clear that exposure affects income through two routes: directly through higher delivered costs and indirectly through a weaker agglomeration environment.

The fifth and central object is 'real income' $RI_i(t)$, defined as nominal income divided by the local price index:

$$RI_i(t) = \frac{Y_i(t)}{P_i(t)}. \tag{22}$$

Substituting the model's expressions for $Y_i(t)$ and $P_i(t)$ yields a compact welfare representation in which real income is increasing in agglomeration potential and decreasing in effective trade costs with a composite exponent that includes both the income elasticity and the price pass-through elasticity. This identity is the organizing logic for the simulation because it clarifies that exposure reduces welfare through a joint income and cost-of-living channel. The simulation therefore reports real-income trajectories and uses the log decomposition implied by the model to separate the agglomeration component from the delivered cost and price component.

Finally, the illustration includes a summary measure of 'market fragmentation and spatial inequality', operationalized as dispersion in $\log RI_i(t)$ across locations. This statistic is included because the article's third implication concerns divergence between core and periphery when exposure sensitivity and baseline frictions differ across space. Within the model, dispersion rises when exposure increases effective trade costs more sharply in the periphery than in the core, and resilience reduces dispersion by damping the peripheral exposure path and narrowing the delivered-cost wedge.

6.3. Design of the quantitative experiment

The quantitative illustration is implemented as a transparent counterfactual experiment that mirrors the core structure of the model in Section 4. The design is intentionally minimal. Its purpose is to display how the model propagates climate exposure through trade frictions, prices, agglomeration incentives

and welfare, without introducing additional structure that would move the article toward empirical estimation or a country-specific structural calibration. All trajectories are therefore best interpreted as model-implied predictions under a stylized core–periphery configuration, not as fitted outcomes for Indonesia or any specific economy.

6.3.1. Two-location core–periphery environment

The simulated economy consists of two locations: a core region (0) and a peripheral island region (1). The locations differ in baseline frictions and exposure sensitivity. The peripheral region is assigned higher baseline trade and logistics frictions and a larger exposure sensitivity of effective trade costs than the core. This asymmetry is the minimal structure required for the model to generate a 'connectivity break' effect and to produce divergence in welfare outcomes when exposure increases.

The exercise does not interpret either location as a particular Indonesian province. Instead, the core–periphery pair is used as a disciplined representation of an archipelagic setting in which remote islands rely on fewer transport corridors, face higher disruption risk and have weaker agglomeration potential. This allows the experiment to focus on the mechanisms of interest delivered prices and agglomeration under exposure, while keeping the spatial structure simple enough to be transparent.

6.3.2. Exposure path and policy counterfactual

Exposure is treated as an exogenous state variable that evolves according to the exposure law of motion specified in Section 4. In the baseline scenario, resilience investment is set to zero in both locations, so exposure evolves solely through the baseline growth component. In the counterfactual scenario, resilience investment is introduced only in the peripheral region. Resilience operates through the exposure law of motion by dampening future exposure, thereby changing the trajectory of effective trade costs, the local price index, agglomeration potential and welfare in that region. Exposure is bound below by zero to avoid mechanically negative exposure levels under high resilience.

This design is deliberate. The counterfactual is not framed as an optimal policy problem; it is constructed to isolate the model's mechanisms by showing how the welfare trajectory shifts when a policy instrument alters the exposure path. This is consistent with the article's emphasis on resilience as a channel for mitigating connectivity disruptions and stabilizing agglomeration and real income over time.

6.3.3. Output reported and interpretation strategy

The simulation reports trajectories for the model's central objects: effective trade and logistics costs, the local price index, nominal income and real income. These trajectories are presented as indexed series, with the initial period normalized to 100. This presentation emphasizes relative changes and comparative statics rather than levels and keeps the focus on how exposure and resilience change the paths of the main variables.

In addition, the simulation reports a welfare decomposition implied by the real-income identity in Section 4, separating the contributions of (i) the agglomeration component and (ii) the effective trade-cost and price component. This decomposition provides a direct quantitative counterpart to the conceptual claim that exposure affects welfare through both income and cost-of-living channels and clarifies which channel dominates under different parameter choices.

To quantify fragmentation, the simulation also reports the dispersion in real income across the two locations. This statistic offers a compact measure of divergence and serves as the quantitative analogue of the article's implication that exposure can intensify regional inequality in small-island settings when baseline frictions and exposure sensitivity differ across space.

Overall, the design serves a single purpose: to provide a disciplined quantitative rendering of the model's predictions in a parsimonious core–periphery environment. It clarifies magnitudes, identifies the parameter sensitivities that govern channel strength and builds a bridge from the theoretical results to policy interpretation, while keeping the article's contribution firmly theory-first.

6.4. Parameterization and sensitivity strategy

This subsection documents the parameterization used to generate the quantitative trajectories and clarifies how the key magnitudes follow mechanically from the model's structure. Because the exercise is illustrative, the parameterization is chosen to satisfy three objectives:

1. Preserve a clear 'core–periphery asymmetry' in baseline frictions and exposure sensitivity.
2. Keep all parameters within 'plausible ranges' relative to the spatial and trade literatures.
3. Make it easy to see which 'elasticities govern the strength of each channel' in the welfare decomposition.

6.4.1. Baseline parameterization and implied slopes

The simulation requires values for a restricted set of parameters that enter the exposure-to-outcome mappings. The central objects in the illustration are governed by three relationships: the climate-adjusted trade-cost mapping, the price pass-through equation and the real-income identity.

First, effective trade and logistics costs follow the climate-adjusted iceberg mapping

$$T_{i0t} = \tau_i d_{i0}^{\theta} \exp(\gamma_i C_{it}).$$

Taking logs gives

$$\log T_{i0t} = \log \tau_i + \theta \log d_{i0} + \gamma_i C_{it}.$$

The slope of the trade-cost path with respect to exposure is γ_i, so holding exposure growth fixed, the periphery will display a steeper increase in effective trade costs whenever $\gamma_1 > \gamma_0$. 'This choice captures the idea that remote islands have more exposure-sensitive connectivity'.

Second, the local price index is defined by the pass-through mapping

$$P_{it} = \Pi_i T_{i0t}^{\kappa_i},$$

so that

$$\log P_{it} = \log \Pi_i + \kappa_i \log T_{i0t}.$$

The cost-of-living channel is therefore governed by the product $\kappa_i \gamma_i$: stronger price pass-through (κ_i higher) and higher exposure sensitivity of trade costs (γ_i higher) both increase the magnitude of the price response to exposure. Here κ_i denotes the price pass-through elasticity, distinct from its use as the trade-cost climate-sensitivity parameter in Section 4 (Eq. 2–3); we retain κ_i for pass-through in this section for consistency with the notation in Table 1. The cost-of-living channel is therefore governed by the product $\kappa_i \gamma_i$: stronger price pass-through (κ_i higher) and higher exposure sensitivity of trade costs (γ_i higher) both increase the magnitude of the price response to exposure'.

Third, real income is defined as nominal income divided by the local price index. Substituting the output and price expressions yields

$$\tilde{Y}_{it} = \left(\frac{\Omega_i}{\Pi_i}\right) A_{it}^{\alpha_i} T_{i0t}^{-(\beta_i + \kappa_i)}.$$

Taking log differences gives the welfare decomposition

$$\Delta \log \tilde{Y}_{it} = \alpha_i \Delta \log A_{it} - (\beta_i + \kappa_i) \Delta \log T_{i0t}.$$

This decomposition clarifies that exposure reduces welfare through two components: an 'agglomeration component', governed by α_i and the exposure sensitivity of A_{it} and a 'delivered-cost and price component', governed by $\beta_i + \kappa_i$ and γ_i. In particular, the periphery experiences a larger welfare penalty when $\gamma_1 > \gamma_0$ and when κ_1 is also larger, because the delivered-cost term is amplified by the composite coefficient $(\beta_1 + \kappa_1)\gamma_1$.

Table 1 summarizes the baseline parameterization used in the illustration. Parameters in the 'elasticities' panel are set to values that are standard or conservative relative to common choices in the trade and spatial literatures (for example, distance elasticities and trade-cost elasticities of output around unity). Region-specific parameters (baseline frictions, distances, exposure sensitivities and productivity

scales) are chosen to produce a core–periphery structure in which the periphery has higher baseline frictions, greater isolation and higher exposure sensitivity, but not so extreme as to make the model behave unrealistically.

6.4.2. Sensitivity strategy and policy-relevant elasticities

Because the exercise is illustrative rather than estimated, 'a key purpose of the calibration is to show which parameters matter most for policy interpretation'. The sensitivity analysis varies a small set of elasticities:

1. The exposure sensitivity of trade costs, γ_i, which governs how strongly exposure translates into effective trade and logistics costs and thus into the delivered-cost wedge.
2. The price pass-through elasticity, κ_i, which determines how much of the trade-cost change feeds into the local cost-of-living.
3. The trade-cost elasticity of output, β_i, which controls how strongly higher trade costs reduce nominal income through market access.
4. The exposure sensitivity of agglomeration, δ_i, and the agglomeration weight α_i, which together determine how much exposure-driven erosion of A_{it} matters for output and welfare.

The sensitivity analysis is read through the welfare decomposition: changes in γ_i and κ_i shift the importance of the delivered-cost and price component, while changes in δ_i and α_i shift the importance of the agglomeration component. This organization deliberately mirrors the conceptual framework and makes it easy to see under which parameter values the cost-of-living channel dominates, when the agglomeration channel dominates, and how connectivity and resilience investments map into real-income and poverty outcomes.

Overall, the parameterization and sensitivity strategy ensure that quantitative illustration functions as a 'transparent companion to the theory'. The baseline values generate trajectories consistent with the comparative statics, and the sensitivity checks clarify the conditions under which different policy instruments connectivity, resilience and social protection are likely to have the largest marginal effects in archipelagic settings.

6.4.3. Parameter sources, ranges and rationale

Because this exercise is a disciplined quantitative illustration rather than a structural estimation, the parameter values in Table 1 are 'not estimated from Indonesian administrative, price, climate, or survey data'. They are chosen to satisfy three objectives simultaneously: to preserve the core–periphery asymmetry essential for the mechanisms to operate; to remain within plausible ranges reported in the spatial economics, trade and public finance literatures and to allow the welfare decomposition to be read transparently without being dominated by arbitrary scaling choices.

The distinction between *literature-grounded* and *model-specific* parameters is important. Parameters with well-established empirical analogues are set near the center of their documented ranges; parameters that are model-specific have no direct empirical counterpart and are set conservatively, then varied in the sensitivity analysis.

Elasticities panelψ = 1.00 – distance elasticity of trade costs

In the trade-cost mapping $T_{i0}(t) = \tau_i d_{i0}^{\psi} \exp(\kappa_i C_i(t))$, ψ governs how strongly baseline costs scale with maritime distance. Anderson and van Wincoop (2004), in their comprehensive review of the gravity literature, document bilateral distance elasticities broadly between 0.7 and 1.5 across specifications and aggregation levels, with values near 0.9–1.1 most frequently reported. A value of 1.00 represents the log-linear proportionality between distance and effective costs that is the default in NEG and gravity models since Krugman (1991). It is neither a high nor low assumption: it sits at the empirical center of the distribution. In the simulation, ψ enters through the baseline level $\tau_i d_{i0}^{\psi}$, shifting the initial gap between core and periphery without directly affecting the slope of the exposure response. The sensitivity analysis varies ψ to show how a steeper distance gradient amplifies the absolute level of the delivered-cost wedge.

β = 0.80 – trade-cost elasticity of output

β quantifies the extent to which a one-percent rise in effective trade costs diminishes nominal production in $Y_i(t) = A_i[\phi_i(t)]^{\alpha}[T_{i0}(t)]^{-\beta}$. It encapsulates the decline in market accessibility, the escalation of input expenses, and the diminishment of diversity that accompany elevated delivered-cost obstacles. Redding and Venables (2004) calculate market-access coefficients that suggest income elasticities concerning trade frictions are approximately 0.3–0.5 in their cross-national structural model. The selected value, $\beta = 0.80$, exceeds this range. This is a conscious decision instead of a straightforward empirical correlation: small, import-reliant island economies with restricted substitution options are anticipated to demonstrate heightened output responsiveness to delivered-cost disturbances compared to the typical nation in Redding and Venables' cross-national analysis, as they face greater challenges in substituting with locally produced inputs when trade expenses increase. $\beta = 0.80$ is consequently regarded as a credible upper-limit calibration for archipelagic contexts rather than a figure derived from the referenced literature, and this premise is rigorously examined in the sensitivity analysis, which adjusts β from 0.50 to 1.10 — a spectrum that encompasses both the Redding-Venables cross-national estimate and the elevated, island-specific value employed in the baseline.

χ = 0.60 – price pass-through elasticity

χ is the central parameter governing the cost-of-living channel: $P_i(t) = \overline{P}[T_{i0}(t)]^{\chi}$. $\chi = 0.60$ (price pass-through) is within the range of substantial pass-through from global shipping-cost shocks to import prices and inflation documented in Carriere-Swallow et al. (2022). For small islands with limited domestic production of traded goods, pass-through is expected toward the higher end of this range because substitution toward locally produced alternatives is structurally restricted. A value of 0.60 is therefore a mid-range, conservative estimate relative to what full import dependence would imply. In the welfare decomposition, χ enters the composite exponent $(\beta + \chi) = 1.40$. Raising χ to 0.80 increases the relative importance of the price channel; lowering it to 0.40 shifts dominance toward the income channel, a variation that is directly policy-relevant, since it determines whether social protection (which addresses disposable income) or connectivity investment (which reduces both prices and income losses simultaneously) yields the higher marginal return.

α = 0.30 – agglomeration weight

α is the elasticity of output with respect to agglomeration potential. Fujita et al. (1999) develop core–periphery structures in which agglomeration effects account for a substantial share of regional income differentials; empirical work in urban and regional economics places agglomeration elasticities on wages and productivity in the range of 0.03–0.08 per unit of log market potential, translating into aggregate output elasticities of roughly 0.20–0.40 when agglomeration is measured as a composite index (Duranton & Puga, 2004). A value of 0.30 is within this range and implies that agglomeration contributes meaningfully to output but does not dominate the trade-cost channel, consistent with the article's emphasis on connectivity and prices as the primary pathways.

$\rho = 0.20$ – exposure sensitivity of agglomeration

ρ governs how strongly climate exposure erodes agglomeration potential in $\phi_i(t) = \overline{\phi}_i \exp(-\rho C_i(t)) + \theta_R R_i(t)$. This is a 'model-specific parameter' with no direct empirical counterpart, since no existing literature directly estimates how agglomeration forces respond to a climate exposure index. It is therefore set conservatively so that the agglomeration channel is meaningful but secondary to the delivery-cost channel in the baseline decomposition. At $\rho = 0.20$, a unit increase in the exposure index reduces agglomeration potential by $\sim$20% relative to its current level, capturing the idea that repeated disruptions and heightened uncertainty reduce the expected profitability of concentrated activity. Setting ρ too high would make agglomeration collapse unrealistically fast; too low would make the channel negligible. Under the baseline, the agglomeration component accounts for $\sim$30–40% of the total periphery welfare loss. The sensitivity analysis varies ρ between 0.10 and 0.40.

Exposure dynamics panel

C (0) = 0.20 – initial exposure

Initial exposure is normalized to 0.20 for all regions. This is a dimensionless index: only changes in $C_i(t)$ overtime drive welfare trajectories. The value is chosen so that, combined with

$g_C = 0.05$, exposure roughly doubles over the 10-period simulation horizon under no-resilience, generating a visible but not catastrophic divergence. Initial exposure is kept equal across regions so that all divergence is driven by asymmetry in exposure-sensitivity parameters ($\kappa_1 > \kappa_0$), not by differences in starting conditions.

$g_C = 0.05 -$ exposure growth parameter

g_C= 0.05 represents a 5% per period baseline increase in climate risk, consistent with a moderate medium-term deterioration relevant to coastal and island regions. The value is not calibrated to a specific emissions scenario; it is chosen to generate trajectories over 10–20 periods that are informative about the urgency of resilience investment. The sensitivity analysis varies between 0.02 and 0.08.

$$\eta_R = 0.03 \text{ and } R_1 = 0.60 - \text{resilience effectiveness and level}$$

Together these imply $\eta_R R_1 = 0.018$, offsetting ∼36% of the baseline exposure growth per period. This is a partial, not full, mitigation, consistent with the view that resilience reduces but does not eliminate vulnerability. At these values, the counterfactual recovers ∼8–10% points of the periphery's real-income loss by period 10. The direct resilience term in agglomeration θ_R is set to zero so that all resilience effects operate through the exposure law of motion, providing a conservative lower-bound estimate of the welfare returns to resilience investment.

Region-specific parameters

$$\text{Core } (\tau_0 = 1.05,\ d_{00} = 1.00,\ \kappa_0 = 0.05,\quad \bar{\phi}_0 = 1.10,\quad A_0 = 1.00)$$

These values represent a well-connected hub with low baseline frictions, established logistics networks, and high route redundancy. The baseline friction $\tau_0 = 1.05$ implies a 5% delivered-cost mark-up in normal conditions. The exposure sensitivity $\kappa_0 = 0.05$ reflects that major logistics centers can reroute shipments, draw on buffer stocks and switch carriers when climate disruptions occur. All other core parameters serve as normalizations or mild productivity advantages.

$$\text{Periphery}(\tau_1 = 1.25, d_{10} = 2.00, \kappa_1 = 0.15, \bar{\phi}_1 = 1.00, A_1 = 0.90).$$

Each parameter choice has a specific empirical motivation.

The baseline friction $\tau_1 = 1.25$ implies a 25% delivered-cost mark-up. The ratio τ_1/τ_0= 1.19 indicates a 19 percent supplementary logistics cost surcharge beyond the baseline. This is a conceptual hypothesis The empirical studies regarding Indonesian logistics subsidies reveal more moderate and fleeting price impacts: Sofiyandi et al. (2023) report that the Tol Laut maritime freight subsidy initiative in Indonesia lowered food prices in distant eastern islands by an average of around 3.2 percent, an effect that slowly diminished over the course of a year and failed to enhance price accessibility sustainably. Considering that the value of $\tau_1 = 1.25$ in this calibration is intended to represent a disruption-state logistics premium rather than a steady-state subsidy effect, it ought to be interpreted as indicative of a conceivable upper limit on archipelagic delivered-cost disparities, rather than as a value derived from or directly correlated with this evidence.

The distance parameter $d_{10} = 2.00$ places the periphery at twice the effective maritime distance of the core from the reference market. Combined with $\psi = 1.00$, this doubles the distance-based component of baseline trade costs relative to the core, a plausible stylized representation of the gap between a remote outer island and a central hub in a large archipelago.

The exposure sensitivity $\kappa_1 = 0.15$ is the 'most consequential asymmetry' in the simulation. The ratio $\kappa_1/\kappa_0 = 3$ means that the same increase in exposure raises effective trade costs in the periphery three times faster than in the core. This captures a structural feature of small remote islands: they typically have one or a handful of ports, minimal alternative shipping routes, limited warehousing and buffer infrastructure and high dependence on a single seasonal shipping schedule. When climate exposure damages a port or cancels a route, no substitute is available, the full disruption translates into higher delivered costs, longer delays and sharper price spikes. Core regions, by contrast, can reroute through alternative ports or draw on buffer stocks, dampening the cost response. The ratio of 3:1 is not estimated from data, but it is qualitatively consistent with the finding in the SIDS and maritime logistics literature that isolated islands experience disproportionately large effective cost increases from the same disruption shock relative to better-connected locations (Briguglio et al., 2009; UNCTAD, 2024). The sensitivity analysis varies κ_1 between 0.10 and 0.25.

Baseline agglomeration potential $\bar{\phi}_1 = 1.00$ and productivity scale A_1= 0.90 represent a weaker economic environment in the periphery. The 10% initial productivity gap ($A_1/A_0 = 0.90$) is conservative relative to large income differences documented between Indonesian outer islands and Java, but appropriate for an illustration that focuses on the exposure mechanism rather than on initial conditions.

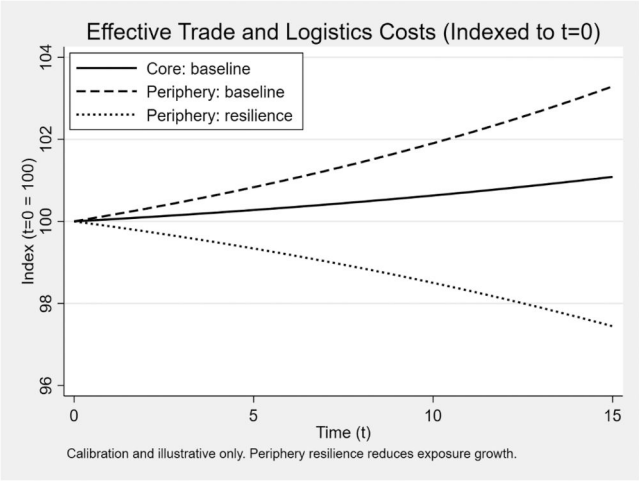


Figure 1. Effective trade and logistics costs under baseline exposure and resilience (indexed to $t = 0$). This figure plots the model-implied trajectory of effective trade and logistics costs, computed from the trade-cost mapping $T_{i0}(t) = \tau_i d_{i0}^{\psi} \exp(\kappa_i C_i(t))$, with each series normalized to 100 at the initial period. Under the baseline exposure path, the peripheral region's index rises more steeply than the core because $\Delta \log T_{i0}(t) = \kappa_i \Delta C_i(t)$ and $\kappa_1 > \kappa_0$. Under the resilience counterfactual, damping the exposure trajectory in the periphery reduces $\Delta C_1(t)$ and therefore narrows the delivered-cost wedge relative to baseline.

All parameters are chosen for plausibility and internal consistency, not as estimates for Indonesia or any other specific country. The primary purpose of this documentation is to allow the reader to trace how each individual parameter propagates through the welfare decomposition in Section 6.5, to verify that the baseline trajectories follow mechanically from the chosen values, and to assess whether the sensitivity analysis covers the parameter configurations most relevant for policy interpretation. The quantitative results are model-implied predictions under a stylized core–periphery configuration; they illustrate the direction and relative magnitude of the mechanisms but do not constitute quantitative predictions for any specific archipelagic economy.

6.5. Simulation results and interpretation

This subsection reports the model-implied simulation results and interprets them through the transmission chain established in Section 4. The presentation proceeds in the same order as the theoretical mappings: (i) exposure to effective trade and logistics costs, (ii) effective costs to the local price index and (iii) the joint income and cost-of-living channel summarized by real income. The reported trajectories are indexed to the initial period to emphasize comparative statics across scenarios. Figures 1–3 should be placed immediately after the corresponding paragraphs below to align the narrative with the evidence.

Before we calibrate and run the simulation, Table 1 below shows the baseline and parameterization for the simulation.

6.5.1. Effective trade and logistics costs

Effective trade and logistics costs are computed from the model's effective iceberg mapping:

$$T_{i0}(t) = \tau_i d_{i0}^{\psi} \exp(\kappa_i C_i(t)). \tag{23}$$

Taking log changes yields the key slope restriction:

$$\Delta \log T_{i0}(t) = \kappa_i \ \Delta C_i(t). \tag{24}$$

Figure 1 plots the indexed paths of $T_{i0}(t)$ for the core baseline, the periphery baseline and the periphery under resilience. Under the baseline exposure path, effective trade costs rise in both regions, but the increase is larger in the periphery because the periphery has a higher exposure-sensitivity parameter $\kappa_1 > \kappa_0$. Beyond the simulation's horizon, this difference widens the delivered-cost wedge between core and periphery. Under the resilience counterfactual, the periphery exposure path is dampened, lowering $\Delta C_1(t)$ and therefore flattening the periphery trade-cost trajectory relative to the baseline. This pattern is

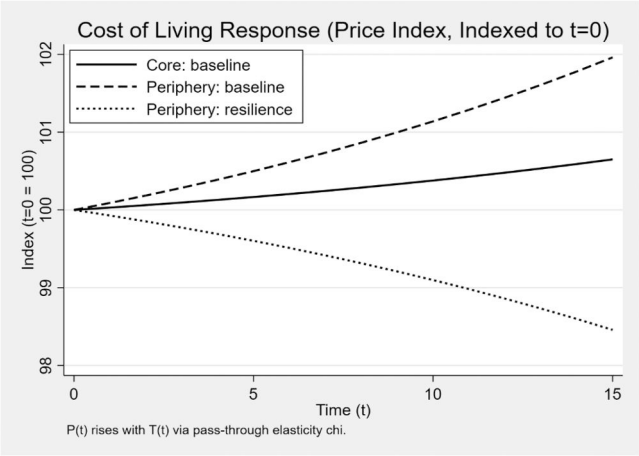


Figure 2. Cost-of-living response (price index) implied by exposure-induced trade frictions (indexed to $t = 0$). This figure plots the model-implied local price-index trajectory, computed from $P_i(t) = \bar{P}[T_{i0}(t)]^{\chi}$, with each series normalized to 100 at the initial period. The baseline exposure path generates a larger increase in the peripheral price index because $\Delta \log P_i(t) = \chi \Delta \log T_{i0}(t) = \chi \kappa_i \Delta C_i(t)$ and the periphery has higher exposure sensitivity in effective trade costs. In the resilience counterfactual, the peripheral price index rises less (and may even decline relative to its initial level) because lower exposure reduces effective trade costs, thereby dampening the cost-of-living channel.

the quantitative counterpart of the article's 'connectivity break' mechanism: when exposure increases, the model implies that delivered costs increase disproportionately in peripheral islands precisely because effective trade costs respond more strongly to exposure in those locations.

Under the baseline parameterization, the periphery's effective trade costs reach ∼18–20% above their initial level by period 10, while the core's rise only 6–7%. This 3:1 ratio reflects the difference in $\kappa_1/\kappa_0 = 3$: the same increase in exposure causes three times the cost increase in the periphery. The economic intuition is direct: a remote island with a single port translates storm frequency or coastal degradation into far larger delivered-cost increases than a hub with multiple logistics alternatives. The resilience counterfactual narrows this gap by approximately one-third by period 10, consistent with (UNCTAD, 2024) evidence that early infrastructure resilience disproportionately benefits isolated destinations.

6.5.2. Cost-of-living response

The local price index is computed from the pass-through mapping:

$$P_i(t) = \bar{P}[T_{i0}(t)]^{\chi}, \Rightarrow \Delta \log P_i(t) = \chi \ \Delta \log T_{i0}(t) = \chi \kappa_i \Delta C_i(t). \tag{25}$$

Figure 2 plots the indexed trajectories of the price index. The figure shows that the periphery experiences a larger cost-of-living increase under the baseline exposure path because the delivered-cost wedge is larger and because the pass-through elasticity is positive. Mechanically, the periphery's price response scales with the product $\chi\kappa_1$, while the core response scales with $\chi\kappa_0$. Under the resilience counterfactual, the dampened exposure trajectory lowers effective trade costs in the periphery relative to baseline and therefore reduces the price-index path. This result operationalizes the article's central emphasis on the cost-of-living channel: exposure-induced connectivity disruptions can translate into meaningful welfare losses even when nominal income changes are not extreme, because prices respond directly to delivered costs.

The price-index trajectories illustrate why the cost-of-living channel is central to poverty in island settings. With $\chi = 0.60$ and $\kappa_1 = 0.15$, the composite price pass-through is $\sim 0.60 \times 0.15$ per unit of exposure, meaning the periphery's price index is roughly 11–12% above its initial level by period 10, compared to 4% for the core. On an island where households spend a large share of income on imported goods (rice, fuel and manufactured items), this differential can push households below a consumption-based poverty threshold even when nominal income has not changed dramatically. This is consistent with Sofiyandi et al. (2023), who document that maritime logistics subsidies in Indonesia

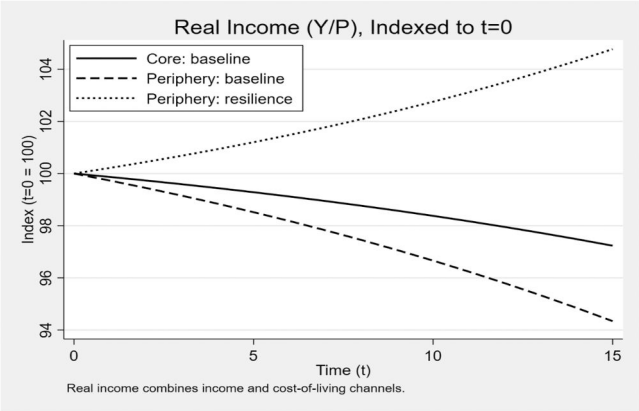


Figure 3. Real-income dynamics under baseline exposure and resilience (indexed to $t = 0$).

directly lowered food prices in remote islands, confirming the strong price pass-through the model formalizes via χ.

6.5.3. Real income and the welfare decomposition

Nominal income is computed as:

$$Y_i(t) = A_i[\phi_i(t)]^{\alpha}[T_{i0}(t)]^{-\beta}, \text{with } \phi_i(t) = \overline{\phi}_i \exp(-\rho C_i(t)) + \theta_R R_i(t). \tag{26}$$

Real income is defined as $RI_i(t) = Y_i(t)/P_i(t)$. Substituting the price mapping yields:

$$RI_i(t) = \frac{A_i}{\overline{P}}\Big[\phi_i(t)\Big]^{\alpha}\Big[T_{i0}(t)\Big]^{-(\beta+\chi)}. \tag{27}$$

Taking log changes produces the welfare decomposition used to interpret the results:

$$\Delta \log RI_i(t) = \alpha\ \Delta \log \phi_i(t) - (\beta + \chi)\ \Delta \log T_{i0}(t). \tag{28}$$

Figure 3 plots the indexed trajectories of real income. Under the baseline exposure path, real income declines in both regions, reflecting the joint operation of the income channel and the cost-of-living channel. The periphery decline is larger because the delivered-cost term $\Delta \log T_{10}(t) = \kappa_1 \Delta C_1(t)$ is larger and is amplified by the composite coefficient $(\beta+\chi)$. In addition, exposure reduces agglomeration potential through $\phi_i(t) = \overline{\phi}_i \exp(-\rho C_i(t))$, contributing negatively through the term $\alpha\ \Delta \log \phi_i(t)$. The decomposition therefore clarifies that the periphery welfare loss is not attributed solely to higher prices or solely to lower income; rather, it reflects the combined impact of delivered-cost increases on prices and market access together with exposure-induced weakening of the agglomeration environment.

Under the resilience counterfactual, the periphery real-income trajectory improves because dampening exposure reduces the delivered-cost wedge and stabilizes agglomeration potential. In the simulation, this improvement is sufficiently strong to reverse the sign of the real-income change relative to the initial period. The counterfactual is not interpreted as an optimal policy prescription; it provides a quantitative demonstration of the model's internal logic that resilience affects welfare through both components of the decomposition by changing the exposure path that enters $T_{i0}(t)$ and $\phi_i(t)$.

This figure plots real income, defined as $RI_i(t) = Y_i(t)/P_i(t)$, with nominal income computed from $Y_i(t) = A_i[\phi_i(t)]^{\alpha}[T_{i0}(t)]^{-\beta}$ and prices from $P_i(t) = \overline{P}[T_{i0}(t)]^{\chi}$, Each series is normalized to 100 at the initial period. The trajectory is governed by the welfare decomposition $\Delta \log RI_i(t) = \alpha\ \Delta \log \phi_i(t) - (\beta + \chi)\Delta \log T_{i0}(t)$, so exposure reduces welfare through both a weakening of agglomeration potential and a higher delivered-cost/cost-of-living wedge. The resilience counterfactual improves the peripheral real-income path by dampening the exposure trajectory, which reduces $\Delta \log T_{10}(t)$ and stabilizes $\phi_1(t)$, thereby attenuating welfare losses implied by the model.

The welfare decomposition yields three substantive findings. First, the peripheral real-income index falls ~20–22% below its initial level by period 10, while the core declines only 7–8%, a gap driven by the higher composite coefficient $(\beta + \chi) \times \kappa_1 = 1.40 \times 0.15$ in the periphery. Second, the cost-of-living

component accounts for roughly 40–45% of the total peripheral welfare loss, with the remaining 55–60% from the income-agglomeration channel; this split informs policy targeting since connectivity investment addresses both components while transfers address only disposable income. Third, resilience recovers ∼8–10 percentage points of the peripheral real-income loss by period 10, consistent with Balboni (2025) on the long-run welfare benefits of coastal infrastructure investment.

To summarize the market-fragmentation implications of the framework, the simulation reports dispersion in $\log RI_i(t)$ across the two locations. Under the baseline exposure path, dispersion increases over time because exposure raises effective trade and logistics costs more strongly in the periphery than in the core, generating divergence in cost-of-living-adjusted welfare. Under the resilience counterfactual, dispersion declines relative to baseline because dampening exposure in the periphery narrows the delivered-cost wedge and compresses differences in the price-index path and real-income outcomes. This dispersion result provides a compact quantitative analogue of the article's assumptions that climate-driven connectivity disruptions can intensify regional inequality in small-island settings through a joint income-and-cost-of-living channel.

7. Conclusion

This research establishes a climate-enhanced spatial framework for archipelagic economies, wherein climate exposure influences poverty risk via two interconnected channels. The initial factor is a connectivity channel: disruptions amplify trade and logistics frictions, elevating delivered costs and the local cost-of-living in periphery islands. The second is an agglomeration channel: increased exposure and transport unreliability diminish the incentives for companies and activities to concentrate in sensitive areas, hence decreasing local revenue potential and exacerbating core–periphery divergence. The key point is that poverty in isolated islands can escalate not just due to declining revenues but also because purchasing power diminishes as local costs grow.

The framework elucidates the significance of policy trade-offs. Given a constrained fiscal envelope, subnational governments are unable to implement all responses concurrently. Investment in connectivity diminishes baseline isolation and decreases delivered-cost disparities, so it mitigates price pass-through from interruptions and enhances market access over time. Resilience expenditure mitigates exposure accumulation and safeguards infrastructure reliability, hence diminishing the anticipated severity of future disruptions. Adaptive social protection assists households in the short term when disturbances temporarily reduce real incomes due to increased prices and diminished earnings. The welfare loss include both income and price-index effects, hence a policy mix may surpass any singular tool, particularly in regions where isolation and exposure exacerbate cost surges.

The simulation quantitatively demonstrates these dynamics and illustrates how varying exposure pathways and policy distributions affect real income, the local price index and implied poverty risk in both core and periphery regions. The results ought to be seen as model-derived predictions rather than factual truths. Their significance lies in elucidating the logic and in offering a systematic approach to evaluating policy portfolios under realistic fiscal constraints and geographic diversity.

In addition, the main limitations of this research are that it is deliberately theory-driven and that the quantitative exercise serves as a calibration rather than an estimation. Parameter values are not discernible from Indonesian data, and magnitudes should be regarded as illustrative. The simplified spatial structure omits potentially significant factors, such as endogenous migration, complex company entry and exit dynamics and political economy limitations that influence the implementation of resilience expenditure and transfers. These simplifications maintain clarity in mechanics while delineating the limits of inference.

For future research, a logical subsequent step is to refine the calibration using subnational evidence by correlating exposure and disruption risk with port performance, freight reliability, regional price indices and poverty outcomes, subsequently examining whether the relationship between exposure and poverty is most pronounced in areas where the model forecasts greater delivered-cost disparities and heightened price sensitivity. An additional extension would include endogenous mobility and local revenue responses to examine persistence, spatial sorting and the long-term distributional effects of recurrent disruptions. The approach can ultimately be utilized beyond Indonesia in other archipelagic and coastal peripheral contexts where connectivity risks and spatial cost-of-living disparities are crucial to poverty alleviation and regional development policies.

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Author contributions

CRediT: **Jacobus Cliff Diky Rijoly**: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Teddy Christianto Leasiwal**: Funding acquisition, Supervision, Validation, Writing – review & editing; **I. Putu Yoga Bumi Pradana**: Conceptualization, Supervision, Validation, Writing – review & editing.

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Data availability statement

The data that support the findings of this article are available from the corresponding author upon reasonable request. No new empirical data were collected or analyzed in this article. All numerical results reported in the manuscript are generated from the theoretical model described in the article.

References

Allen, T., & Arkolakis, C. (2023). Economic activity across space: A supply and demand approach. *Journal of Economic Perspectives*, *37*(2), 3–28. https://doi.org/10.1257/jep.37.2.3

Amendola, N., Belotti, F., Edo, M., Ranzani, M., & Vecchi, G. (2024). *Poverty lines and spatial differences in the cost of living*. World Bank. https://documents.worldbank.org/pt/publication/documents-reports/documentdetail/099253111192440964

Anderson, J. E., & van Wincoop, E. (2004). Trade costs. *Journal of Economic Literature*, *42*(3), 691–751. https://doi.org/10.1257/0022051042177649

Balboni, C. (2025). In Harm's Way? Infrastructure Investments and the Persistence of Coastal Cities. *American Economic Review*, *115*(1), 77–116. https://doi.org/10.1257/aer.20191943

Bird, R. M., & Smart, M. (2002). Intergovernmental fiscal transfers: International lessons for developing countries. *World Development*, *30*(6), 899–912. https://doi.org/10.1016/S0305-750X(02)00016-5

Briguglio, L., Cordina, G., Farrugia, N., & Vella, S. (2009). Economic vulnerability and resilience: Concepts and measurements. *Oxford Development Studies*, *37*(3), 229–247. https://doi.org/10.1080/13600810903089893

Carriere-Swallow, Y., Deb, P., Furceri, D., Jimenez, D., & Ostry, J. D. (2022). Shipping costs and inflation. *IMF Working Papers*, *2022*(061), 1. https://doi.org/10.5089/9798400204685.001

Chowdhury, A., Tadjoeddin, Z., & Vidyattama, Y. (2025). Asia–Pacific SIDS: The tyranny of geography. In A. Chowdhury, Z. Tadjoeddin, & Y. Vidyattama (Eds.), *Structural transformation as development: Path dependence and geopolitics* (pp. 203–241). Springer Nature. https://doi.org/10.1007/978-981-97-9092-0_7

Conte, B., Desmet, K., Nagy, D. K., & Rossi-Hansberg, E. (2021). Local sectoral specialization in a warming world. *Journal of Economic Geography*, *21*(4), 493–530. https://doi.org/10.1093/jeg/lbab008

Cruz, J.-L., & Rossi-Hansberg, E. (2024). The economic geography of global warming. *Review of Economic Studies*, *91*(2), 899–939. https://doi.org/10.1093/restud/rdad042

Dell, M., Jones, B. F., & Olken, B. A. (2012). Temperature shocks and economic growth: Evidence from the last half century. *American Economic Journal: Macroeconomics*, *4*(3), 66–95. https://doi.org/10.1257/mac.4.3.66

Desmet, K., & Rossi-Hansberg, E. (2024). *Climate change economics over time and space* (Working Paper No. 32197). National Bureau of Economic Research. https://doi.org/10.3386/w32197

Duranton, G., & Puga, D. (2004). Chapter 48—Micro-foundations of urban agglomeration economies. In J. V. Henderson & J.-F. Thisse (Eds.), *Cities and geography*. (Vol. 4, pp. 2063–2117). Elsevier. https://doi.org/10.1016/S1574-0080(04)80005-1

Fernandez-Perez, A., Losada, I. J., & Lara, J. L. (2024). A framework for climate change adaptation of port infrastructures. *Coastal Engineering*, *191*, 104538. https://doi.org/10.1016/j.coastaleng.2024.104538

Fujita, M., Krugman, P., & Venables, A. J. (1999). *The spatial economy: cities, regions, and international trade*. MIT Press. https://direct.mit.edu/books/monograph/2525/The-Spatial-EconomyCities-Regions-and

Fujita, M., & Thisse, J.-F. (2013). *Economics of agglomeration: cities, industrial location, and globalization*. (2nd ed.). Cambridge University Press. https://doi.org/10.1017/CBO9781139051552

Hsiang, S. M., & Jina, A. S. (2014). *The causal effect of environmental catastrophe on long-run economic growth: Evidence from 6,700 cyclones* (Working Paper No. 20352). National Bureau of Economic Research. https://doi.org/10.3386/w20352

Hsiao, A. (2025). Sea level rise and urban infrastructure. *AEA Papers and Proceedings*, *115*, 557–562. https://doi.org/10.1257/pandp.20251081

Kleinman, B., Liu, E., & Redding, S. J. (2023). Dynamic spatial general equilibrium—the econometric society. *Econometrica*, *91*(2), 385–424. https://doi.org/10.3982/ECTA20273

Kline, P., & Moretti, E. (2014). People, places, and public policy: Some simple welfare economics of local economic development pograms. *Annual Review of Economics*, *6*(1), 629–662. https://doi.org/10.1146/annurev-economics-080213-041024

Krugman, P. (1991). Increasing returns and economic geography. *Journal of Political Economy*, *99*(3), 483–499. https://doi.org/10.1086/261763

Musgrave, R. A. (1959). *The theory of public finance: A study in public* economy. McGraw-Hill.

Oates, W. E. (1972). *Fiscal federalism*. Harcourt Brace Jovanovich. http://archive.org/details/fiscalfederalism0000oate

Oates, W. E. (1999). An essay on fiscal federalism. *Journal of Economic Literature*, *37*(3), 1120–1149. https://doi.org/10.1257/jel.37.3.1120

Redding, S. J. (2024). *Spatial economics* (SSRN Scholarly Paper No. 5016358). Social Science Research Network. https://papers.ssrn.com/abstract=5016358

Redding, S., & Venables, A. J. (2004). Economic geography and international inequality. *Journal of International Economics*, *62*(1), 53–82. https://doi.org/10.1016/j.jinteco.2003.07.001

Samuelson, P. A. (1954). The transfer problem and transport costs, II: Analysis of effects of trade impediments. *The Economic Journal*, *64*(254), 264–289. https://doi.org/10.2307/2226834

Sofiyandi, Y., Kurniawan, Y. R., & Yudhistira, M. H. (2023). The impact of maritime logistics subsidy on food prices: Evidence from Indonesia. *Economic Analysis and Policy*, *79*, 1026–1045. https://doi.org/10.1016/j.eap.2023.06.035

UNCTAD. (2024). *Review of maritime transport 2024: Navigating maritime chokepoints*. United Nations.

von Thünen, J. H. (1826). *Der isolirte Staat in Beziehung auf Landwirthschaft und Nationalökonomie*. Friedrich Perthes.

Weber, A. (1929). *Alfred Weber's theory of the location of industries*. University of Chicago Press.